

M U L 2 _ N E W

Theoretical Guide

Carrera Unified Formulation, finite elements and numerical methods
behind MUL2_NEW

Politecnico di Torino · MUL2 project
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Introduction and scope

1.1 What this guide is for

This guide explains the **theory** behind MUL2_NEW, a finite-element program for the linear analysis of beams, plates, shells-as-plates and solids written with the *Carrera Unified Formulation* (CUF). It is written for a student or a new developer who knows basic mechanics of solids and has seen the finite element method once, and who wants to understand *what the program computes and why*, before looking at *how it is coded* (the Implementation Guide) or *how it is used* (the User Guide).

The three guides are meant to be read together:

Guide	Question it answers	Typical reader
Theoretical Guide (this one)	What equations are solved? Which approximations are made?	Anyone who wants to trust or extend the results
Implementation Guide	How are those equations coded? Where is each formula?	Developers
User Guide	How do I describe a structure and run it?	Analysts and students

1.2 What the program does

MUL2_NEW solves two problems for linear elastic structures:

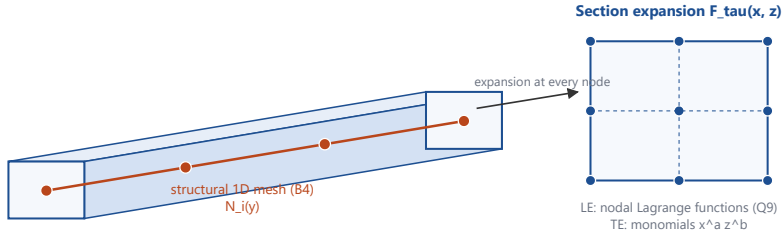
- **Analysis 101 – linear static analysis.** Given a structure, supports and point loads, find the displacement field u that satisfies equilibrium, and recover strains and stresses.
- **Analysis 103 – linear free vibration.** Find the natural frequencies f_k and the mode shapes ϕ_k of the unloaded, supported structure.

Both are *mechanical* problems: only the three displacement fields u , v , w are unknowns. (The unified formulation has room for temperature, electric potential, humidity and mixed stress fields, and the input syntax already contains the corresponding columns; they are not active in 101/103.)

The distinguishing feature of CUF is that **beam, plate and solid models are produced by the same code**. A structure is described by

1. a *structural mesh* – one-dimensional (beam), two-dimensional (plate) or three-dimensional (solid) finite elements; and
2. an *expansion over the remaining dimensions* – for a beam the cross-section, for a plate the thickness – chosen freely: a polynomial (Taylor, TE) or a finite-element mesh of the section (Lagrange, LE).

Changing the expansion changes the kinematic theory (from an Euler-Bernoulli-like beam to a three-dimensional-quality solution) *without changing a line of code*. This is why a few thousand lines of Fortran cover a very wide family of models.



$$\mathbf{u}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \sum_i \sum_{\tau} \mathbf{N}_i(\mathbf{y}) \mathbf{F}_{\tau}(\mathbf{x}, \mathbf{z}) \mathbf{u}_{\tau, i}$$

Figure 1.1. Beam as a one-dimensional structural mesh whose nodes carry a section expansion.

1.3 Notation

Vectors and matrices are written in bold, scalars in *italic*. Indices i, j number structural nodes, τ, s number expansion terms, p a structural Gauss point and q an expansion Gauss point. The conventions used throughout (and in the code) are:

Table 1.2. Notation used in the three guides.

Symbol	Meaning	Code name
x, y, z	local element coordinates; for a beam y is the axis and x - z the section; for a plate x - y is the mid-surface and z the thickness	LOCAL frame
$\mathbf{u} = (u, v, w)$	displacement	SOLUTION
ϵ	strain $[\epsilon_{xx}, \epsilon_{yy}, \epsilon_{zz}, \gamma_{xz}, \gamma_{yz}, \gamma_{xy}]^T$ (engineering shear strains)	STRAIN_*
σ	stress $[\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{xz}, \sigma_{yz}, \sigma_{xy}]^T$	STRESS_*
$\mathcal{C}$	6×6 stiffness (elasticity) matrix	STIFFNESS_LOCAL
$N_i(\xi)$	structural shape function of node i	SHAPE
$F_{\tau}(x, z)$	expansion function number τ	EPS , FACTOR_VALUE
K, M	global stiffness and mass matrices	STIFFNESS , MASS
ξ, η, ν	natural coordinates of an element	NATURAL
J	Jacobian of the isoparametric map	JACOBIAN

NOTE

The engineering-shear convention matters. The shear strains $\gamma_{xz}, \gamma_{yz}, \gamma_{xy}$ are *twice* the tensor components. Stress and strain therefore transform differently under rotation (Chapter 5), and the constitutive matrix couples them without extra factors of two.

1.4 Structure of the guide

Chapter 2 recalls the continuum mechanics and the finite element ideas that are needed. Chapter 3 introduces the element library. Chapters 4 to 6 develop the unified formulation, the reference systems and the constitutive models. Chapter 7 derives the element matrices, including the *separable* form that makes high-order expansions affordable. Chapter 8 treats shear locking, Chapter 9 loads and constraints, Chapters 10 and 11 the solution of the static and eigenvalue problems, Chapter 12 the recovery of strains and stresses and Chapter 13 the verification of the program. Appendices collect numerical tables and references.

Mechanics and finite element background

2.1 Linear elasticity in matrix form

A deformable body occupies a volume Ω with boundary $\partial\Omega$. Under small displacements the strain is the symmetric gradient of the displacement. With the engineering-shear ordering of Chapter 1 it is written with a differential operator D :

$$\epsilon = D u, \quad D u = \begin{bmatrix} \partial_x u \\ \partial_y v \\ \partial_z w \\ \partial_z u + \partial_x w \\ \partial_z v + \partial_y w \\ \partial_y u + \partial_x v \end{bmatrix} \quad (2.1)$$

Hooke's law gives the stress as a linear function of the strain through the 6×6 symmetric positive-definite matrix C :

$$\sigma = C \epsilon \quad (2.2)$$

2.2 Principle of virtual displacements

Equilibrium of a body loaded by body forces b , surface tractions and point forces F_k is expressed weakly: for every admissible virtual displacement δu (zero where displacements are prescribed),

$$\int_{\Omega} \delta \epsilon^T \sigma \, d\Omega = \int_{\Omega} \delta u^T b \, d\Omega + \sum_k \delta u(x_k)^T F_k. \quad (2.3)$$

For free vibration there are no loads and the inertia term $-\rho \ddot{u}$ is added to the internal work. Looking for harmonic motion $u(x) e^{i\omega t}$,

$$\int_{\Omega} \delta \epsilon^T \sigma \, d\Omega = \omega^2 \int_{\Omega} \rho \delta u^T u \, d\Omega. \quad (2.4)$$

Everything that follows is the discretisation of (2.3) (static) and (2.4) (modal).

2.3 The finite element method in one page

Choose a finite-dimensional space of displacement fields

$$u(x) = N(x) q,$$

where q collects the unknown coefficients ("degrees of freedom", DOF) and N the interpolation functions. Then $\epsilon = DNq = Bq$ and the weak forms become matrix equations:

$$Kq = F, \quad K\phi = \omega^2 M\phi, \quad K = \int_{\Omega} B^T C B \, d\Omega, \quad M = \int_{\Omega} \rho N^T N \, d\Omega. \quad (2.5)$$

K is the *stiffness* matrix, M the *consistent mass* matrix. The integrals are computed element by element and the element matrices are summed (*assembled*) into global matrices.

2.4 Isoparametric elements

An element is described on a *reference domain* with natural coordinates ξ , for instance $[-1, 1]^2$ for a quadrilateral. The same shape functions $N_i(\xi)$ interpolate both the geometry and the unknown:

$$x(\xi) = \sum_i N_i(\xi) x_i. \quad (2.6)$$

The Jacobian $J = \partial x / \partial \xi$ links derivatives and volumes:

$$\frac{\partial N_i}{\partial x} = J^{-T} \frac{\partial N_i}{\partial \xi}, \quad d\Omega = \det J \, d\xi. \quad (2.7)$$

In MUL2_NEW the Jacobian of a structural element is built in the **element local frame** and only along the axes that the element really spans (the axis y for a beam, the plane x - y for a plate). This is explained in Chapter 5.

2.5 Numerical integration

Integrals over the reference element are computed by Gauss–Legendre quadrature,

$$\int_{-1}^1 f(\xi) \, d\xi \approx \sum_{p=1}^n w_p f(\xi_p),$$

which is *exact* for polynomials of degree $2n - 1$. Tensor products give rules for quadrilaterals and hexahedra. The default number of points per direction is the one that integrates the product of two shape-function derivatives exactly on an undistorted element:

Table 2.1. Default Gauss rules.

Element	Points per direction	Total points
B2, Q4, H8	2	2, 4, 8
B3, Q9, H20, H27	3	3, 9, 27
B4, Q16	4	4, 16
T3	1 point at the centroid	1
T6	3 points (mid-edge rule)	3

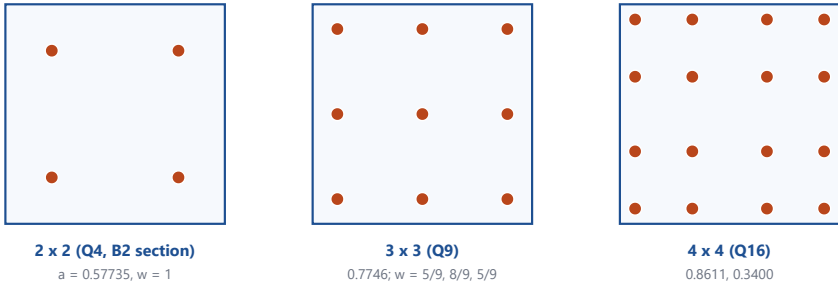


Figure 2.1. Gauss-Legendre points of the quadrilateral rules.

WARNING

The default rule is **not** enough for the expansion of a Taylor model of high order. A Taylor expansion of order N contains monomials up to degree N and the stiffness integrand contains products of two of them, i.e. degree $2N$ in each direction. $N + 1$ Gauss points per direction are then required. MUL2_NEW selects this number automatically for the expansion meshes (Section 7.6).

2.6 Sparse matrices and DOF numbering

A global matrix has one row and one column per DOF but each DOF is coupled only with the DOFs of the elements that share its node. The matrix is therefore *sparse*. MUL2_NEW stores it in *compressed sparse row* (CSR) format (see the Implementation Guide). The order in which DOFs are numbered does not change the mathematics but fixes the layout of the output files: the **numbering is field-major** (all u DOFs, then all v , then all w).

The element library

MUL2_NEW offers fourteen structural topologies plus a one-node "point" element used as the degenerate expansion of solids. The same topologies are used for the *expansion meshes* (sections and thicknesses) that carry the Lagrange expansion.

Table 3.1. Element topologies.

Name	Nodes	Natural dim.	Role	Shape functions	Default Gauss rule	MITC
B2	2	1	beam	linear Lagrange	2	yes
B3	3	1	beam	quadratic Lagrange	3	yes
B4	4	1	beam	cubic Lagrange	4	yes
Q4	4	2	plate / section	bilinear	2×2	yes
Q9	9	2	plate / section	biquadratic	3×3	yes
Q16	16	2	plate / section	bicubic	4×4	no
T3 (Q3)	3	2	plate / section	linear triangle	1 point	no
T6 (Q6)	6	2	plate / section	quadratic triangle	3 points	no
H8	8	3	solid	trilinear	2×2×2	yes
H20	20	3	solid	serendipity (geometry only)	3×3×3	yes
H27	27	3	solid	triquadratic	3×3×3	yes
T4, T10, P6	4, 10, 6	3	solid	tetra / prism	—	no
S1	1	0	expansion of solids	constant	1	—

NOTE

Shape functions of H20, T4, T10 and P6 are listed in the input syntax and the topology registry but are not yet implemented in the shape-function module of the current release; selecting them gives an explicit error. Q3/Q6 are accepted as aliases of T3/T6.

3.1 Node numbering

The order of the nodes in the input record is the order of the shape functions. It must follow the conventions below; a different order produces a distorted or inverted element.

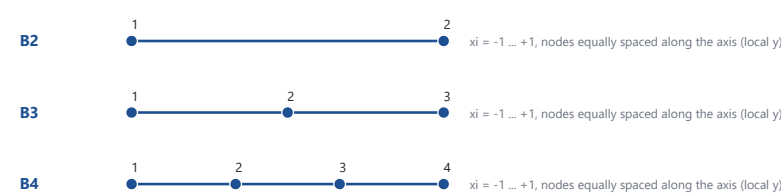


Figure 3.1. Beam elements: nodes are numbered along the axis.

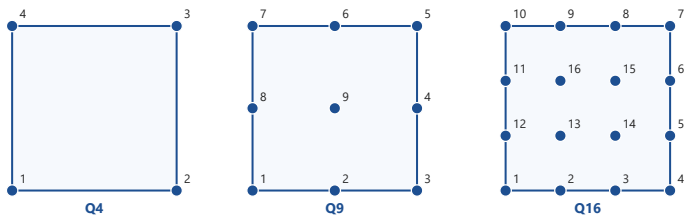


Figure 3.2. Quadrilaterals: corners and mid-side nodes alternate counter-clockwise, interior nodes last.

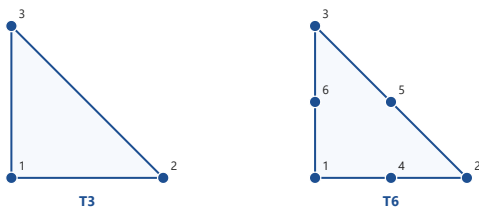
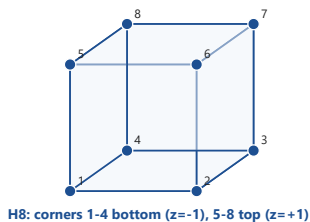


Figure 3.3. Triangles.



H27 node classes

1-8 corners (same order as H8)

9-20 mid-edge nodes

21-26 mid-face nodes

27 centre node

Natural coordinates (ξ, η, ν) in $[-1, 1]^3$.

H20 has corners and mid-edge nodes only.

P6: two triangles (1-3 and 4-6); T4/T10: tetrahedra.

Figure 3.4. Hexahedra.

3.2 One-dimensional Lagrange functions

The line elements use equally spaced nodes on $[-1, 1]$. For order m (number of nodes $m + 1$) the function of node a is

$$\ell_a^m(\xi) = \prod_{b \neq a} \frac{\xi - \xi_b}{\xi_a - \xi_b}, \quad \xi_b = -1 + \frac{2(b-1)}{m}. \quad (3.1)$$

Quadrilaterals and hexahedra are tensor products: for the biquadratic quadrilateral (Q9) node 7, for instance, sits at (ξ_1, ξ_3) of the 3×3 grid and $N_7 = \ell_1^2(\xi) \ell_3^2(\eta)$. The nodal order is given by the integer tables (IX, IV, IZ) in the shape-function module and follows Figure Figure 3.2.

The triangles use area coordinates $L_1 = 1 - \xi - \eta$, $L_2 = \xi$, $L_3 = \eta$:

$$\text{T3: } N_a = L_a, \quad \text{T6: } N_{1,2,3} = L_a(2L_a - 1), \quad N_4 = 4L_1L_2, \quad N_5 = 4L_2L_3, \quad N_6 = 4L_3L_1.$$

3.3 Geometry in the local frame

Each structural element gets a **right-handed local frame** (x, y, z) built from its node coordinates and a user-supplied reference vector (Chapter 5). The geometric map and its Jacobian are evaluated **in that frame and only along the directions spanned by the element**:

- beam: one coordinate y (the axis); $J = [\partial y / \partial \xi]$ is a scalar;
- plate: two coordinates (x, y) in the mid-surface; J is 2×2 ;
- solid: three coordinates; J is 3×3 .

The gradient of a structural shape function is stored as a 3-vector in the local frame with zero components along the axes the element does not span. This is the object that enters the strain operator in Chapter 4.6.

The *section* (beam) or *thickness* (plate) is **not** part of the structural Jacobian: it has its own Jacobian, computed from the expansion mesh nodes (Chapter 4). The total volume element is the product

$$d\Omega = \det J_s \det J_e d\xi_s d\xi_e, \quad (3.2)$$

where s stands for the structural and e for the expansion factor. This product structure is what the *separable kernel* of Chapter 7 exploits.

The Carrera Unified Formulation

4.1 The idea

Classical beam theories (Euler–Bernoulli, Timoshenko) and plate theories (Kirchhoff, Reissner–Mindlin) postulate how the displacement varies across the section or thickness: linearly, with a rotation, and so on. The unified formulation does not postulate a single law. It writes the displacement as a **sum of products** of a function of the structural coordinate and a function of the remaining coordinates, and lets the user choose the second family:

$$u(x, y, z) = \sum_{\tau=1}^T F_{\tau}(x, z) u_{\tau}(y) \quad (\text{beam}) \quad (4.1)$$

where the *generalised displacements* $u_{\tau}(y)$ are functions of the axis only and are themselves discretised with beam finite elements:

$$u_{\tau}(y) = \sum_{i=1}^{N_n} N_i(y) u_{\tau i}.$$

The complete approximation is therefore

$$u_k(x, y, z) = \sum_{i=1}^{N_n} \sum_{\tau=1}^T N_i(y) F_{\tau}(x, z) u_{k\tau i}, \quad k \in \left\{ u, v, w \right\} \quad (4.2)$$

For a plate the structural functions N_i depend on (x, y) and the expansion $F_{\tau}(z)$ on the thickness coordinate only; for a solid the expansion is the constant $F_1 = 1$ (the "S1" element) and the formulation reduces to the standard three-dimensional finite element method.

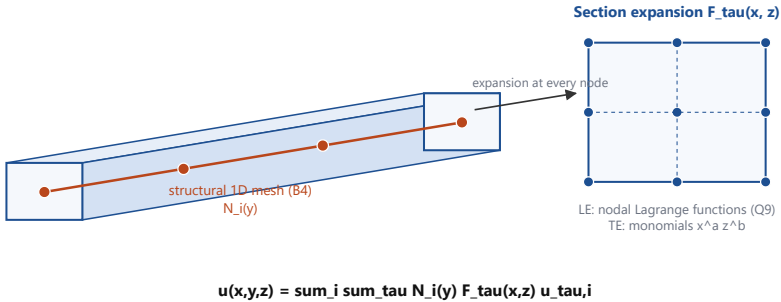


Figure 4.1. Structural functions N_i (red) times expansion functions F_{τ} (blue).

Why this is powerful. The code never needs to know which theory it implements: it loops over structural nodes i and expansion terms τ and multiplies two scalar functions and their derivatives. The *kinematic theory* is determined entirely by the list of functions F_{τ} and by their number T .

4.2 Field-dependent kinematics

The three components u, v, w need not use the same expansion. A node has a *kinematic* record that assigns an expansion family and order to each of the nine fields of the formulation (U, V, W, T, P, B, SZZ, SXZ, SYZ). In analyses 101/103 only U, V, W may be active. With the historical `NODES.dat` the three components of a node share the same expansion; with a `KINEMATICS.dat` file they may differ; and the kernels allow it (the strain column of a DOF is always built from the scalar basis of **its own** field).

Different nodes may also use different kinematics. At an interface between a high-order node and a low-order node the unknowns of the missing terms are simply absent; compatibility is then guaranteed only if the lower expansion is a subset of the higher one (e.g. TE2 next to TE4). A mixed LE/TE interface is not conforming in general.

4.3 Taylor expansions (TE)

In a Taylor expansion the functions F_τ are the monomials of the section coordinates up to order N :

$$F_\tau(x, z) = x^a z^b, \quad a + b \leq N. \quad (4.3)$$

The terms are ordered by increasing total degree and, inside a degree, by increasing power of z : 1; x , z ; x^2 , xz , z^2 ; ... The number of terms is

$$T_N^{\text{beam}} = \frac{(N+1)(N+2)}{2}, \quad T_N^{\text{plate}} = N+1.$$

Taylor monomials $x^a z^b$ of a section, order $N = 0..4$

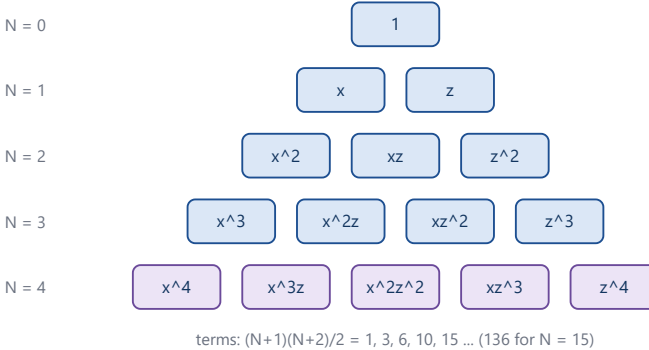


Figure 4.2. Taylor monomials of a section.

Some orders are classical models in disguise:

Table 4.1. Classical models contained in the Taylor family (beam, displacement approach).

Order	Terms	Displacement across the section	Corresponds to
TE0	1	rigid translation	membrane/axial model without section kinematics
TE1	3	linear in x, z	Timoshenko-type beam (rotations and warping-free section)
TE2	6	quadratic	enriched beam with in-plane deformation and Poisson effect
TE3, TE4 ...	10, 15, ...	cubic, quartic ...	refined models for thick, short or non-symmetric sections

WARNING

TE0 is not a classical beam: without section terms the Poisson contraction cannot develop and a pure axial test with $\nu \neq 0$ shows spurious stiffening. Use TE1 or higher for engineering results. The uniform-strain patch tests of the verification suite use $\nu = 0$ for this reason.

Conditioning. Monomials of high order on a small section are badly scaled: on a 10×1 mm section, $z^{15} \approx 10^{-45}$ while $1 \approx 1$. The stiffness entries then span many orders of magnitude and the matrix becomes numerically singular. Practical remedies are to model the section in units close to one (for instance metres $\rightarrow$ millimetres) or to use Lagrange expansions, which are well conditioned for any size. The program prints a warning when it has to retry the factorization with a more robust algorithm (Chapter 10).

4.4 Lagrange expansions (LE)

In a Lagrange expansion the section is itself a small finite element mesh (the *expansion mesh*) of Q4, Q9, Q16 or triangular elements (or line elements B2, B3, B4 for the thickness of a plate). Each node τ of the expansion mesh carries a Lagrange function F_τ , equal to one at node τ and zero at all the other nodes. The unknown $u_{k\tau i}$ is then **the physical displacement of the point of the section that sits at expansion node τ** , at structural node i . There is no rotation-like variable: the "rotations" are implied by the differences of the displacements of different points of the same section.

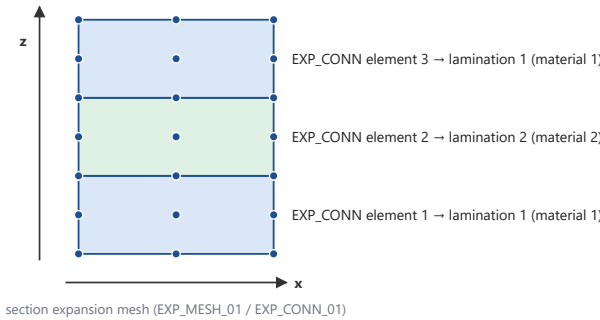


Figure 4.3. A laminated section: each ply is one sub-element with its own material.

Two properties make LE the natural tool for composites and local analysis:

1. The expansion is **layer-wise**: a sub-element belongs to one lamination, so the constitutive matrix jumps between sub-elements while the displacement stays continuous (interlaminar displacement continuity is automatic).
2. The functions have **local support**: terms τ and s belonging to expansion nodes that do not share a sub-element never couple in the stiffness or mass matrix. The program exploits this to leave those entries out of the sparse pattern (Implementation Guide, Chapter 7).

The *order* of the Lagrange family is not an input: it is determined by the topology of the expansion elements (Q4 $\rightarrow$ linear; Q9 $\rightarrow$ quadratic; Q16 $\rightarrow$ cubic).

4.5 Degrees of freedom

The number of unknowns is

$$n_{\text{DOF}} = \sum_{\text{nodes}} \sum_{k \in \{u, v, w\}} T_k(\text{node}).$$

Table 4.2. DOF per structural node for a few common models (three displacement fields).

Model	Terms T	DOF per node
Beam, TE1	3	9
Beam, TE2	6	18
Beam, TE4	15	45
Beam, TE15	136	408
Beam, LE with 4 Q9 sub-elements (25 nodes)	25	75
Plate, TE2 through the thickness	3	9
Plate, LE with 3 B2 layers (4 nodes)	4	12
Solid (H8, H27)	1	3

4.6 Strain and the fundamental nucleus

Inserting (4.2) in the strain operator (2.1) gives a strain that is a linear combination of the *derivatives of the products* $N_i F_\tau$. Writing $\phi_{i\tau} = N_i F_\tau$ and using the local frame (x, y, z) ,

$$\nabla \phi_{i\tau} = \begin{bmatrix} \partial_x \phi \\ \partial_y \phi \\ \partial_z \phi \end{bmatrix} = \begin{bmatrix} N_i \partial_x F_\tau \\ \partial_y N_i F_\tau \\ N_i \partial_z F_\tau \end{bmatrix} \quad (\text{beam}) \quad (4.4)$$

while for a plate the structural gradient has components along x, y and the expansion gradient along z . The strain produced by one unit coefficient of component u_k is the column

$$b_{k i \tau} = B(\nabla \phi_{i\tau}) e_k,$$

where $B(g)$ is the 6×3 operator that maps a displacement of *unit value times* a scalar function with gradient g into strains:

$$B(g) = \begin{bmatrix} g_x & 0 & 0 \\ 0 & g_y & 0 \\ 0 & 0 & g_z \\ g_z & 0 & g_x \\ 0 & g_z & g_y \\ g_y & g_x & 0 \end{bmatrix}. \quad (4.5)$$

The element stiffness is then the sum, over pairs of generalised displacements, of integrals of $b^T C b$. The integral of one such pair is the **fundamental nucleus** of the formulation. In a hand-written CUF code the nucleus is a 3×3 block whose entries are combinations of four *section integrals* (of $F_\tau F_s$, $F_{\tau,x} F_s$, $F_\tau F_{s,x}$, $F_{\tau,x} F_{s,x}$, ...) and of the corresponding structural integrals. MUL2_NEW recovers exactly this structure automatically from the scalar bases (Chapter 7).

4.7 Choosing the expansion: a decision table

Table 4.3. Which expansion for which problem.

Situation	Recommended	Notes
Slender isotropic beam, global response	TE1–TE2	cheapest; TE1 needs v -effects only through TE2
Thick beam, short beam, local stress concentration	TE3–TE5 or LE	LE with moderate mesh is better conditioned
Thin-walled or open section	LE with a fine section mesh	warping and distortion are captured by the mesh
Laminated plate or sandwich	LE through the thickness (B2/B3 per ply or per layer group)	layer-wise displacement continuity
Thick plate	TE2–TE4 through the thickness	
Fully three-dimensional	H8/H27	the S1 expansion is the identity

Reference systems

Every computation in the program is done in a well defined **local frame** of each element; the results are then rotated to the global frame on request. Mixing up frames is the most common source of wrong models, so this chapter is precise about them.

5.1 The three frames

1. **Global frame** (X, Y, Z) : the frame of the node coordinates, of the loads, of the constraints and (by default) of the output.
2. **Element frame** (x, y, z) : right-handed, built for every element from its geometry and a reference vector (the *versor*). Strains, stresses and the constitutive matrix are evaluated in this frame.
3. **Material frame** (T, L, Z) : the principal axes of the material (fibre direction L , transverse T , thickness Z). It is related to the element frame by two angles per lamination.

The sequence of transformations is



5.2 Element frame

Let $R \in \mathbb{R}^{3 \times 3}$ be the matrix whose *rows* are the local axes expressed in global components. A vector with global components a has local components $a_{loc} = Ra$, and conversely $a = R^T a_{loc}$. (In the code `GLOBAL_TO_LOCAL` is R and `LOCAL_TO_GLOBAL` its transpose.)

5.2.1 Solids

For three-dimensional elements the element frame is the global frame: $R = I$. The versor in the connectivity record is ignored.

5.2.2 Beams

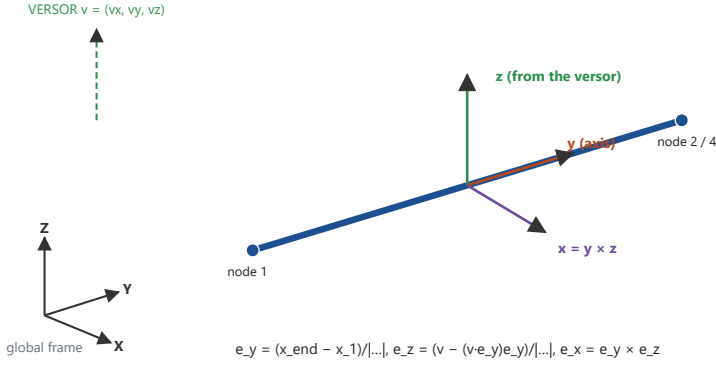


Figure 5.1. Beam frame: y along the axis, z from the versor.

With p_1 the first and p_e the last node of the element ($e = 2, 3, 4$ for B2, B3, B4) and v the versor;

$$e_y = \frac{p_e - p_1}{\|p_e - p_1\|}, \quad e_z = \frac{v - (v \cdot e_y) e_y}{\| \cdot \|}, \quad e_x = e_y \times e_z. \quad (5.1)$$

So the **beam axis is the local y** , the versor chooses the direction of the local z (up to its component along the axis, which is removed) and x completes the right-handed triad. The section coordinates of the expansion mesh are measured on x and z ; the node of the expansion mesh with coordinates $(x_s, 0, z_s)$ is physically located at

$$X = X_{node} + R^T (x_s, 0, z_s)^T. \quad (5.2)$$

NOTE

If the versor is parallel to the beam axis the element frame is not defined and the program stops with the error `SOR IS PARALLEL TO BEAM AXIS`.

Example. A beam along the global Y axis with versor $(0, 0, 1)$ has $e_y = (0, 1, 0)$, $e_z = (0, 0, 1)$ and $e_x = e_y \times e_z = (1, 0, 0)$: local and global axes coincide. A beam along X with versor $(0, 0, 1)$ has $e_y = (1, 0, 0)$, $e_z = (0, 0, 1)$, $e_x = (0, -1, 0)$.

5.2.3 Plates

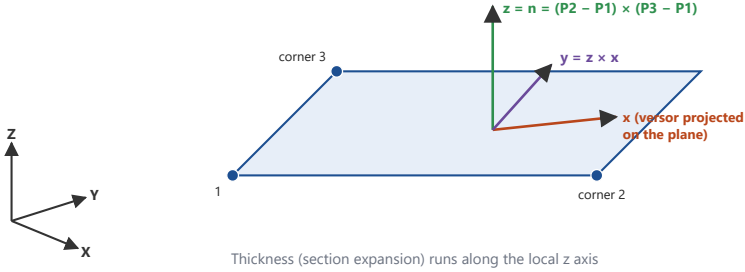


Figure 5.2. Plate frame: z normal to the surface.

With p_1 the first node, p_2 and p_3 two further corners (for Q4 nodes 2 and 4, for Q9 nodes 3 and 7, for Q16 nodes 4 and 10, for triangles nodes 2 and 3),

$$e_z = \frac{(p_2 - p_1) \times (p_3 - p_1)}{\| \cdot \|}, \quad e_x = \frac{v - (v \cdot e_z)e_z}{\| \cdot \|}, \quad e_y = e_z \times e_x. \quad (5.3)$$

The **plate normal is the local z** and the thickness expansion lives along it. The versor selects the local x direction as its projection on the plate. The ordering of the nodes therefore decides which side is "up".

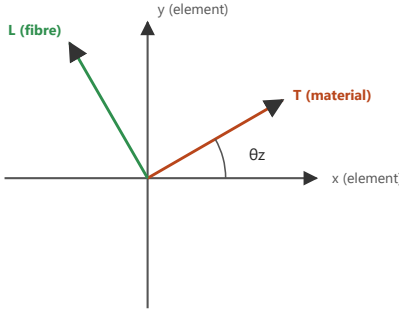
WARNING

A plate element whose three corner nodes are collinear, or a versor perpendicular to the surface, gives the errors DEGENERATE SURFACE OR SUR IS NORMAL TO SURFACE.

5.3 Material frame and lamination angles

The lamination record gives two angles θ_y, θ_z (degrees). The rotation that transforms *element-frame components into material-frame components* is

$$A = R_z(\theta_z) R_y(\theta_y), \quad R_y = \begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix}, \quad R_z = \begin{bmatrix} \cos\theta_z & \sin\theta_z & 0 \\ -\sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (5.4)$$



Material axes (T, L, Z)

LAMINATION.dat gives two angles per lamination
 ANGLE_Y: rotation about the element y axis
 ANGLE_Z: rotation about the element z axis
 $A = R_z(\theta_z) \cdot R_y(\theta_y)$ maps element-frame components to material-frame components.
 With $\theta_y = 0$ and $\theta_z = 0$ the material axes
 T, L, Z coincide with the element x, y, z.

Figure 5.3. Material axes of a lamina rotated by θ_z about the element z axis.

With $\theta_y = \theta_z = 0$ the material axes coincide with the element axes: $T \equiv x$, $L \equiv y$, $Z \equiv z$. For a unidirectional composite beam with fibres along its axis, L is the axis (local y) and the angles are zero. For a ply of a laminated plate whose fibres lie at 45° to the local x direction, $\theta_z = 45^\circ$ (about the plate normal z).

5.4 Transformation of strains and stresses

Because the shear strains are engineering shears, strain and stress transform with *different* 6×6 matrices. With the Voigt order (xx, yy, zz, xz, yz, xy), for a rotation A (a_{ij} its entries) the strain transformation T_ϵ is

$$T_\epsilon = \begin{bmatrix} a_{11}^2 & a_{12}^2 & a_{13}^2 & a_{13}a_{11} & a_{12}a_{13} & a_{12}a_{11} \\ a_{21}^2 & a_{22}^2 & a_{23}^2 & a_{23}a_{21} & a_{23}a_{22} & a_{22}a_{21} \\ a_{31}^2 & a_{32}^2 & a_{33}^2 & a_{33}a_{31} & a_{33}a_{32} & a_{32}a_{31} \\ 2a_{31}a_{11} & 2a_{32}a_{12} & 2a_{33}a_{13} & a_{33}a_{11} + a_{31}a_{13} & a_{33}a_{12} + a_{32}a_{13} & a_{31}a_{12} + a_{32}a_{11} \\ 2a_{31}a_{21} & 2a_{32}a_{22} & 2a_{33}a_{23} & a_{33}a_{21} + a_{31}a_{23} & a_{33}a_{22} + a_{32}a_{23} & a_{31}a_{22} + a_{32}a_{21} \\ 2a_{21}a_{11} & 2a_{12}a_{22} & 2a_{13}a_{23} & a_{13}a_{21} + a_{11}a_{23} & a_{13}a_{22} + a_{12}a_{23} & a_{11}a_{22} + a_{12}a_{21} \end{bmatrix} \quad (5.5)$$

so that $\epsilon_{mat} = T_\epsilon \epsilon_{el}$. The stiffness in the element frame follows from invariance of the strain energy:

$$C_{el} = T_\epsilon^T C_{mat} T_\epsilon. \quad (5.6)$$

The inverse transformations are used for the output: strains and stresses are computed in the element frame and converted to global components on request (GLB) or kept in the element frame (LOC). The stress transformation T_σ is the transpose-inverse pattern of T_ϵ ; since the frames are orthogonal, $T_\epsilon^{-T} = T_\sigma$.

5.5 Displacement components

The displacement unknowns u_{kti} , $k = 1, 2, 3$ are the components of the displacement **in the global frame**. The strain operator is, however, written in the local frame, so the strain column of a global component k uses the local components of the global unit vector e_k :

$$b_{kti} = B(\nabla_{loc} \phi_{it}) (R e_k), \quad (5.7)$$

where $R e_k$ is the k -th column of R . This is why loads, constraints and the displacements in the output are all in the global frame regardless of how the element frames are oriented.

Constitutive models

6.1 The stiffness matrix

All materials are linear elastic. The stress-strain relation is (2.2) with a 6×6 symmetric positive-definite matrix C expressed in the **material axes** (T, L, Z) and in the Voigt order (TT, LL, ZZ, TZ, LZ, TL) , which coincides with (xx, yy, zz, xz, yz, xy) when the material axes coincide with the element axes.

6.2 Isotropic material (ISO-M)

With Young's modulus E , Poisson's ratio ν and the Lamé constants

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad G = \frac{E}{2(1+\nu)},$$

the matrix is

$$C = \begin{bmatrix} \lambda + 2G & \lambda & \lambda & & & \\ \lambda & \lambda + 2G & \lambda & & & \\ \lambda & \lambda & \lambda + 2G & & & \\ & & & G & & \\ & & & & G & \\ & & & & & G \end{bmatrix}. \quad (6.1)$$

The program requires $E > 0$, $-1 < \nu < 0.5$ and $\rho \geq 0$.

6.3 Orthotropic material (ORT-M)

An orthotropic material has three orthogonal symmetry planes. It is defined by nine engineering constants in the input order

$$E_L, E_T, E_Z, \nu_{LT}, \nu_{LZ}, \nu_{TZ}, G_{LT}, G_{LZ}, G_{TZ}, \rho.$$

Internally the axes are ordered $(T, L, Z) = (1, 2, 3)$ and the normal part of the compliance matrix is

$$S_n = \begin{bmatrix} 1/E_T & -\nu_{LT}/E_L & -\nu_{TZ}/E_T \\ -\nu_{LT}/E_L & 1/E_L & -\nu_{LZ}/E_L \\ -\nu_{TZ}/E_T & -\nu_{LZ}/E_L & 1/E_Z \end{bmatrix}, \quad C_n = S_n^{-1}, \quad (6.2)$$

while the shear part is diagonal with $C_{44} = G_{TZ}$, $C_{55} = G_{LZ}$, $C_{66} = G_{LT}$. The program checks that the compliance is positive and non-singular.

NOTE

The definition of the Poisson ratios follows the convention $\nu_{LT} = -\epsilon_T / \epsilon_L$ in a uniaxial test along L . A unidirectional carbon/epoxy lamina with fibres along L typically has $E_L \approx 140$ GPa, $E_T = E_Z \approx 10$ GPa, $\nu_{LT} \approx 0.3$, $G_{LT} \approx 5$ GPa.

6.4 General anisotropic material (MAT-C)

The user can supply the 36 entries of C in the material axes (row by row) followed by the density. The reader checks symmetry and keeps the data even if it is not symmetric; the assembled matrices are symmetric only if C is, and the symmetric solvers require it.

6.5 Orientation: from the material to the element frame

Each lamination refers to one material and rotates it with the angles of Chapter 5. The stiffness in the element frame is $C_{el} = T_e^T C_{mat} T_e$, equation (5.6). This matrix is computed **once per lamination** and cached; every Gauss point of a sub-element that uses the lamination reads the same 6×6 block.

A **laminate** is obtained by giving each sub-element of the expansion mesh its own lamination (and hence material and angle). Because the lamination is attached to the *sub-element of the expansion mesh*, it is constant inside the sub-element and may change from one sub-element to the next, which is exactly the situation the *separable kernel* of Chapter 7 requires.

6.6 Density and the mass matrix

The density of the material of the lamination enters the consistent mass matrix (2.5). In an orthotropic material the density is a scalar: no direction dependence.

6.7 Quantities not used in 101/103

Thermal expansion (T-EXP), specific heat (T-SPC), conductivity (T-CON), viscosity (VISCO), piezoelectric, magnetic, hygroscopic and damping records of the historical material file are *recognised and ignored* by this release (a warning states that non-mechanical data were preserved). They remain in the file format so that the same input files can be used by the full program.

Element matrices and the separable kernel

7.1 From the weak form to the element matrices

Take one structural element e_s with N_n nodes and one expansion mesh with T terms per field. The unknown vector of the element contains $N_n \times 3 \times T$ coefficients, ordered **node by node, then field, then term**:

$$q_e = [\underbrace{u_{1,1}, \dots, u_{T,1}}_{\text{field } u} \underbrace{v_{1,1}, \dots}_{\text{field } v} \underbrace{w_{1,1}, \dots}_{\text{field } w} \mid \text{node 2} \mid \dots]$$

(the *global* numbering is field-major instead; the element-local order is the one used inside the kernels and is mapped to the global order by the DOF list). The element stiffness and mass are the integrals (2.5) over the element volume, which by (3.2) factorise into a structural and an expansion integral:

$$K_e = \sum_{e_x} \sum_p \sum_{q \in e_x} w_p w_q \det J_{s,p} \det J_{e,q} B_{pq}^T C_{e_x} B_{pq}, \quad (7.1)$$

where e_x runs over the sub-elements of the expansion mesh, p over the structural Gauss points and q over the Gauss points of the sub-element. C_{e_x} is the (rotated) stiffness of the lamination of e_x .

7.2 The reference ("point-by-point") evaluation

The direct way of computing (7.1) is to build, at every combined point (p, q) , the strain column of every DOF and accumulate $b_I^T C b_J$ for all pairs (I, J) . The number of operations is of order

$$\underbrace{PQ}_{\text{combined points}} \times \frac{n^2}{2} \times 12, \quad n = 3N_n T.$$

For a TE15 beam with B4 elements ($n = 1632$, $P = 4$, $Q = 256$ per sub-element) this is $\approx 10^{10}$ floating-point operations per element and was the dominant cost in the first version of the code. MUL2_NEW keeps this evaluation as a reference (it is exact for any constitutive and quadrature layout) but uses it only as a fallback; Section 7.4 explains the default.

7.3 Structure of the strain column

Inside the element frame the gradient of one basis function is, for any DOF with structural node i and expansion term τ ,

$$\partial_d(N_i F_\tau) = \sigma_d(x_s; i) \epsilon_d(x_e; \tau), \quad d \in \{x, y, z\}, \quad (7.2)$$

with

$$\sigma_d = \begin{cases} \partial_d N_i & d \in S \\ N_i & d \notin S \end{cases}, \quad \epsilon_d = \begin{cases} F_\tau & d \in S \\ \partial_d F_\tau & d \notin S \end{cases}. \quad (7.3)$$

The set S contains the axes spanned by the structural element: $S = \{y\}$ for a beam, $\{x, y\}$ for a plate and $\{x, y, z\}$ for a solid. In words: along a structural direction one differentiates the structural function and keeps the expansion function; along an expansion direction it is the other way round. Since the operator (4.5) is linear in the gradient,

$$B(\nabla\phi) r_k = \sum_d \partial_d \phi w_{d,k}, \quad w_{d,k} = B(e_d) r_k, \quad (7.4)$$

where $r_k = Re_k$ is the local representation of the global unit vector of component k . The six-component vectors $w_{d,k}$ depend only on the element frame and on the field, never on the position.

7.4 The separable kernel

Insert (7.2) and (7.4) into the stiffness integral of one sub-element e_x and exchange sums. For the pair of DOFs $I = (i, \tau, k)$ and $J = (j, s, h)$

$$K_{IJ}^{(e_x)} = \sum_{d,d'} \underbrace{\Gamma_{kh,dd'}^{(e_x)} \sum_p w_p \det J_{s,p} \sigma_d(p; i) \sigma_{d'}(p; j)}_{S_{dd'}(i,j)} \underbrace{\sum_{q \in e_x} w_q \det J_{e,q} \epsilon_d(q; \tau) \epsilon_{d'}(q; s)}_{E_{dd'}^{(e_x)}(\tau, s)}, \quad (7.5)$$

with the scalar coupling coefficient

$$\Gamma_{kh,dd'}^{(e_x)} = w_{d,k}^T C_{e_x} w_{d',h}. \quad (7.6)$$

Equation (7.5) is the fundamental nucleus of the formulation: a 3×3 **combination of products of a structural matrix** $S_{dd'}$ (**size** $N_n \times N_n$) **and an expansion matrix** $E_{dd'}$ (**size** $T \times T$). The mass matrix is simpler:

$$M_{IJ} = \delta_{kh} \sum_{e_x} \rho_{e_x} \left[\sum_p w_p \det J_{s,p} N_i N_j \right] \left[\sum_{q \in e_x} w_q \det J_{e,q} F_\tau F_s \right]. \quad (7.7)$$

The number of operations of (7.5) is, per sub-element,

$$\underbrace{9 P N_n^2}_{S_{dd'}} + \underbrace{9 Q T^2}_{E_{dd'}} + \underbrace{9 n^2 / 2}_{\text{combination}},$$

i.e. **additive** in the two sets of points instead of multiplicative. For the TE15 example the cost drops from $\approx 10^{10}$ to $\approx 5 \cdot 10^7$ operations per element, and the assembly of a 100 000-DOF model from 526 s to 12 s (Benchmark, Chapter 13).

Separable kernel: cost adds instead of multiplying

POINT-BY-POINT for each of P·Q combined points for each DOF pair ($n^2 / 2$) $B_j^i \wedge T C B_j^i$ cost $\approx P \cdot Q \cdot n^2$	SEPARABLE structural: $S m(i, j) = \sum_p w \sigma (P \cdot N^2)$ expansion: $E m(t, s) = \sum_q w \epsilon (Q \cdot T^2)$ combine: $K = \sum \wedge \cdot E m (n^2 \cdot \#e-9)$ cost $\approx P \cdot N^2 + Q \cdot T^2 + n^2$
--	---

Example: beam B4 (N = 4), Taylor order 15 (T = 136, n = 544·3 = 1632), P = 4, Q = 256 per sub-element

point-by-point $\approx 4 \cdot 256 \cdot 1632^2 / 2 \approx 1.4 \cdot 10^9$ nucleus evaluations per element

separable $\approx 9(4 \cdot 16 + 256 \cdot 136^2 + 1632^2 / 2) \approx 5 \cdot 10^7$ products per element

Measured (100 776 DOF, Taylor 15): assembly 526 s → 12 s

Figure 7.1. Cost of the two evaluations.

Validity conditions:

1. C is constant inside each sub-element. This is true for every lamination type implemented (LAM2).
2. The factorisation (7.2) holds, i.e. structural and expansion directions do not mix. This is true for beams (section $\perp$ axis), plates (thickness $\perp$ surface) and solids.
3. The quadrature is a tensor product of a structural rule and an expansion rule, which is how the Gauss points are laid out.

If any condition fails the program falls back to the point-by-point evaluation of the same integral.

7.5 Locality of the Lagrange expansion

For a Lagrange expansion the function F_τ of expansion node τ vanishes identically in every sub-element that does not contain node τ . Hence $E_{dd}^{(e_x)}(\tau, s) = 0$ whenever τ and s do not both belong to e_x , and summing over sub-elements, $K_{IJ} = 0$ unless τ and s share at least one sub-element. The global matrix therefore has **structural zeros** that are excluded from the sparse pattern. For a 4×4 -element Q9 section the number of non-zeros of a 100 000-DOF beam drops from 120 M to 20 M.

7.6 Taylor expansions: why $N + 1$ Gauss points

The expansion functions of a Taylor model of order N have degree N in each of x and z . The integrand of $E_{dd}(\tau, s)$ contains the product of two such functions (degree $2N$, or less after differentiation) times the (constant for an affine section) Jacobian determinant. A Gauss rule with n_g points integrates exactly polynomials of degree $2n_g - 1$, hence

$$2n_g - 1 \geq 2N \quad \Leftrightarrow \quad n_g \geq N + \frac{12}{12} \Rightarrow \quad n_g = N + 1. \quad (7.8)$$

The program computes, for each expansion mesh, the largest Taylor order of the nodes of the elements that use it, and uses a tensor Gauss rule with $\max(n_g^{\text{default}}, N + 1)$ points per direction on the sub-elements. The rule for $n_g > 4$ is computed by Newton iteration on the Legendre polynomials. With fewer points the

stiffness is **not** the Taylor stiffness: the integration is inexact for $N \geq 3$ on a 3×3 rule and the matrix was found numerically singular already for $N = 4$. (This was the cause of a severe error found by the benchmark of Chapter 13 and is the reason for the regression test `beam_te8_unit.`)

7.7 Mapping of the matrix to the global system

The element matrix is dense in its local DOFs. Each local DOF has a global number given by the field-major layout (Chapter 10). The matrices are scattered into the global CSR arrays in a fixed order, which makes the assembled matrix independent of the number of threads.

7.8 Summary of the element computation

1. For every structural node i and field k obtain the number of terms and the expansion specification from the node kinematic.
2. At every structural Gauss point obtain $N_i, \nabla N_i$ (local frame) and $w_p \det J_{s,p}$.
3. At every expansion Gauss point obtain $F_\tau, \nabla F_\tau$ and $w_q \det J_{e,q}$.
4. Form $S_{dd'}, E_{dd'}$ and Γ ; combine according to (7.5) and (7.7).
5. If MITC is active, replace the structural functions of the tied strain rows (Chapter 8) before step 4.

Shear locking and the MITC technique

8.1 The locking phenomenon

A low-order element that must represent a *thin* structure in bending produces spurious transverse shear strain: the displacement field is not rich enough to give zero shear when the element bends. The shear energy, penalised by a large shear stiffness relative to bending, then dominates and the element is far too stiff. This is called **shear locking**; for plates it is controlled by the ratio of thickness to element size and gets worse as the structure gets thinner.

Locking is a property of the discretisation, not of the physics. The remedy of the **MITC** family (Mixed Interpolation of Tensorial Components, Bathe and Dvorkin) is to replace, in the stiffness integral, the shear strains computed from the displacement derivatives by an **assumed strain field**, interpolated from the strains evaluated at a few special *tying points*.

8.2 Assumed strain

Let $\gamma^d(\xi)$ be a strain component computed in the standard way. In the tied form, one writes

$$\mathcal{V}(\xi) = \sum_l M_l(\xi) \gamma(\xi_l), \quad (8.1)$$

where ξ_l are the tying points and M_l the Lagrange polynomials of the tying set (tensor products of one-dimensional Lagrange polynomials). The other strain components are left untouched. In the CUF context the key point is:

NOTE

The tying acts on the **structural** directions only. The expansion point q is the same at the tying points and at the integration point; only the structural functions N_i and ∇N_i are evaluated at ξ_l and interpolated.

In terms of the nucleus of Chapter 7: for a tied row r the structural factor $\sigma_d(p; i)$ is replaced by

$$\tilde{\sigma}_d^{(r)}(p; i) = \sum_l M_l(\xi_p) \sigma_d(\xi_l; i), \quad (8.2)$$

while the expansion factor $\epsilon_d(q; \tau)$ is unchanged. The structural matrices $S_{dd'}$ then exist in several *classes* (plain, tied by set 1, set 2, ...) and the coupling coefficient is split by row class: $\Gamma_{kh, dd'}^{cc} = \sum_{r \in c, r' \in c'} w_{d, k}[r] C_{rr'} w_{d', h}[r']$.

8.3 Tying tables

The table below reproduces the choices of the baseline program ($a = 1/\sqrt{3}$, $b = \sqrt{3/5}$; the rows are $1 = xx, 2 = yy, 3 = zz, 4 = xz, 5 = yz, 6 = xy$).

Table 8.1. Tied strain rows and tying points.

Element	Rows	Tying points (natural coordinates)
---------	------	------------------------------------

Element	Rows	Tying points (natural coordinates)
B2	5, 6	$\{0\}$
B3	5, 6	$\{-a, +a\}$
B4	5, 6	$\{-b, 0, b\}$
Q4	4	$\{0\} \times \{\pm 1\}$
Q4	5	$\{\pm 1\} \times \{0\}$
Q9	1, 4	$\{\pm a\} \times \{-b, 0, b\}$
Q9	2, 5	$\{-b, 0, b\} \times \{\pm a\}$
Q9	6	$\{\pm a\} \times \{\pm a\}$
H8	4, 5, 6	centre point
H20, H27	4, 5, 6	$\{\pm a\}^3$

Q9 tying points for the tied strain rows (xi horizontal, eta vertical)

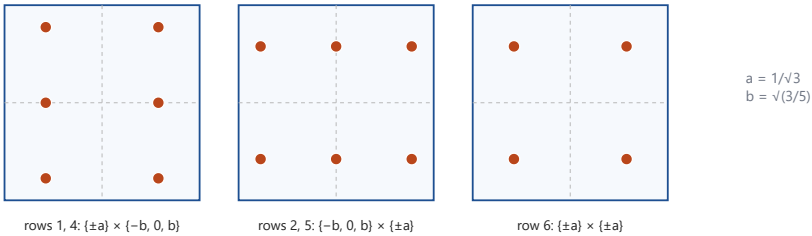


Figure 8.1. Tying points of the Q9 element.

For beams the tied rows are the two transverse shear strains (γ_{xy} and γ_{yz} in the element frame, where y is the axis); the strain of the axis direction is untouched. For a Q4 plate the shear strains γ_{xz} , γ_{yz} are tied; for a Q9 plate the normal and shear in-plane components take part as listed.

8.4 Where MITC is active

The analysis file selects the treatment **per family**: one word for beams, one for plates, one for solids. Values are **NONE** (full integration), **REDI** (reduced integration), **SELI** (selective integration) and **MITC** (tied shear strains).

Topologies without a tying table (Q16, T3, T6, T4, T10, P6) are integrated fully and the program prints a warning. In the validation campaign MITC with the H8 solid was found regular and effective (it reproduces the pure bending exactly), contrary to what an early version of this guide stated.

8.5 Reduced and selective integration

Both act on the integration of the **structural element** (along the axis of a beam, over the surface of a plate, over the volume of a solid); the integration over the section or the thickness is not changed. The reduced rule has one Gauss point per direction less than the full one:

Table 8.2. Gauss points per direction of the structural elements.

Element	B2	B3	B4	Q4	Q9	Q16	H8	H27
full (NONE)	2	3	4	2	3	4	2	3
reduced (REDI , SELI)	1	2	3	1	2	3	1	2

- **REDI**: the whole element, stiffness and mass, is integrated with the reduced rule (as in the baseline program).
- **SELI**: the constitutive matrix is split, $C = C_n + C_s$, where C_s keeps the rows and columns of the strains that cause locking and C_n all the others; C_n is integrated with the full rule, C_s with the reduced one, and the mass with the full rule. The reduced strains are the transverse shears of the element: γ_{yz}, γ_{xy} for beams (the strains that contain the derivative along the axis y), γ_{xz}, γ_{yz} for plates and all three shears for solids.

This is exactly the sum $K = \int B^T C_n B dV|_{full} + \int B^T C_s B dV|_{reduced}$, so the stiffness matrix is symmetric. The two-set implementation reproduces the baseline program to 10^{-7} (displacements and frequencies of beams and plates).

WARNING

with the reduced rule the Q4, H8 and H27 elements keep zero-energy (hourglass) modes: **REDI** can give a singular or meaningless solution for them (the baseline has the same behaviour and the program prints a warning). Use **SELI** or **MITC**. Triangles have no reduced rule and stay fully integrated.

For **REDI** the mass matrix is also reduced and therefore singular; the modal solver detects it and solves $Mx = \mu Kx$, discarding $\mu = 0$ (the infinite frequencies).

8.6 Practical advice

- For *slender beams* with low-order section expansions and B2/B3 elements, MITC is essential; with B4 elements (cubic) locking is already weak.
- For *thin plates* with Q4 or Q9 elements MITC is strongly recommended.
- Locking can also be reduced by using more nodes per element or finer meshes; MITC makes the coarse mesh affordable.
- MITC leaves the stiffness symmetric and positive definite: nothing changes in the solvers.

Loads and constraints

9.1 Point loads

A point load $F = (F_x, F_y, F_z)$ (global components) applied at the physical point X_0 contributes to the right-hand side of the equilibrium equations through the virtual work $\delta u(X_0)^T F$. Since $\delta u_k(X_0) = \sum_{i,\tau} \phi_{i\tau}(X_0) \delta u_{k\tau i}$, the load vector has the entries

$$f_{k\tau i} = F_k N_i(X_0) F_\tau(X_0). \quad (9.1)$$

In practice the program restricts the point to be located *on a structural node and at a point of the section* (this is how the baseline works and what a user normally wants: loads at nodes of the section mesh). Then:

- **Lagrange expansion (LE).** $N_i = 1$ at the structural node and $F_\tau(X_0)$ equals one for the expansion node τ located at X_0 and zero for all others: the load goes **directly to the single DOF** of that node of the section.
- **Taylor expansion (TE).** The functions F_τ are nonzero everywhere, so the load is **projected on all the monomials** evaluated at the section coordinates of the load point: $f_{k\tau i} = F_k X_0^a z_0^b$. A TE model has no node-like quantity: a point load distributes over the terms.

NOTE

In a TE model the result of a point load depends on the evaluated expansion point: loading the section centroid or a corner gives different generalised forces. For a distributed or resultant load on a section use LE or apply several point loads.

If no node of any element matches the load location within tolerance, the program stops with `POINT LOAD LOCATION WAS NOT FOUND`.

9.2 Displacement constraints

The constraint records fix displacement components on a **plane** $AX + BY + CZ + D = 0$ (global coordinates). The set of DOFs affected is determined **geometrically**: for each structural node and each term the physical position

$$X_{i\tau} = X_i + R^T(x_\tau, 0, z_\tau)^T$$

is tested against the plane within a tolerance.

- **LE:** every DOF of a term whose position lies on the plane is constrained ($u_\tau = \bar{u}$). Partial constraints of a section (for example clamping only the bottom face) are possible.
- **TE:** the plane must contain the **whole** expansion of the node (all section points). Then the constant term takes the prescribed value and all the other terms are set to zero. A plane that cuts the section would require a multi-point constraint among the monomials and is rejected with the error `PARTIAL PLANE CONSTRAINT REQUIRES MPC FOR TE`.

Each of the three components can be constrained (to a value) or left free (`N`). Conflicting prescriptions for the same DOF produce an error.

9.3 Exact elimination for the static problem

Let the DOFs be split into free (f) and constrained (c) sets with prescribed values $\bar{q}_c$. The static system

$$\begin{bmatrix} K_{ff} & K_{fc} \\ K_{cf} & K_{cc} \end{bmatrix} \begin{bmatrix} q_f \\ q_c \end{bmatrix} = \begin{bmatrix} F_f \\ F_c \end{bmatrix}$$

is solved for the free part:

$$K_{ff} q_f = F_f - K_{fc} q_c. \quad (9.2)$$

The implementation does not change the size of the matrix: the constrained rows and columns are zeroed, their diagonal is set to one, the right-hand side of the constrained rows is the prescribed value, and the free rows receive the correction $-K_{fc}\bar{q}_c$ (Figure Figure 9.1). The result is mathematically identical to (9.2) and keeps the matrix symmetric and positive definite.

Exact symmetric elimination of constrained DOFs (analysis 101)

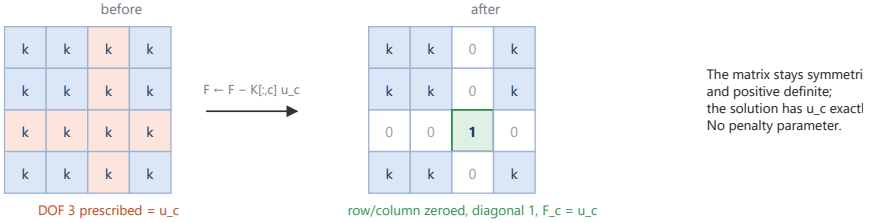


Figure 9.1. Elimination of a constrained DOF.

Comparison with the penalty method. The baseline program imposes constraints by adding a very large stiffness (10^{10}) to the constrained diagonal and a corresponding force. This is simple but (i) perturbs the solution at the level of the ratio of the stiffnesses, (ii) worsens the conditioning, and (iii) produces reaction errors. The exact elimination of MUL2_NEW agrees with the baseline to $\approx 10^{-7}$ for static displacements and $\approx 4 \times 10^{-6}$ for frequencies, which is the order of the penalty perturbation.

9.4 Constraints in the modal problem

For free vibration the constrained DOFs are **removed**: only the free rows and columns of K and M are kept,

$$K_{ff} \phi_f = \omega^2 M_{ff} \phi_f,$$

and the constrained components of the mode are zero. Prescribed non-zero displacements and applied loads have no meaning for a free vibration and are ignored (with a warning).

9.5 Reactions

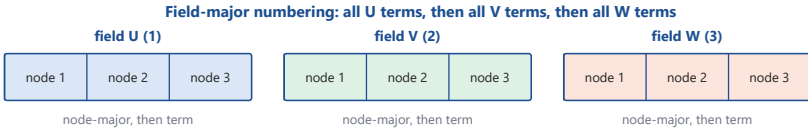
The reaction at a constrained DOF is $R_c = K_{cf} q_f + K_{cc} q_c - F_c$. It is not written by the current release because the constrained rows are modified in place; the dump of K and F (`KMAT`, `FRCE`) can be used to recompute it externally.

Global system and static solution

10.1 Global DOF numbering

Every structural node a and every displacement field $k \in \{u, v, w\}$ owns a block of $T_k(a)$ consecutive DOFs (the number of terms of the expansion of that field at that node). The blocks are numbered **field by field**:

$$\text{DOF}(a, k, \tau) = \text{FIRST}(a, k) + \tau - 1, \quad \text{FIRST}(a, k) = 1 + \sum_{k' < k} \sum_{a'} T_{k'}(a') + \sum_{a' < a} T_k(a'). \quad (10.1)$$



$$\text{global DOF} = \text{FIRST}(\text{node}, \text{field}) + \text{term} - 1$$

TERM_COUNT(node, field): TE order $N \rightarrow (N+1)(N+2)/2$ (beam), $N+1$ (plate), 1 (solid)

LE $\rightarrow$ number of nodes of the expansion mesh

Example: 3 nodes, TE1 beam (3 terms): 9 DOF per field, 27 DOF in total

Figure 10.1. Field-major numbering.

The numbering has no influence on the solution, but it fixes the layout of the dumped matrices and vectors (`K_MAT`, `FORCES`, `UNKNOWN`). Nodes that belong to no element receive no DOF and a warning is issued.

10.2 Sparse storage

Two DOFs are coupled when they appear together in some element. For a Lagrange expansion they must also be associated with expansion nodes that share a sub-element (Chapter 7). The global matrices are stored in *compressed sparse row* format with one-based, 64-bit indices:

- `ROW_POINTER(1:N+1)` : position of the first entry of every row;
- `COLUMN_INDEX(1:nnz)` : column of every entry, ascending in each row, the diagonal always present;
- `STIFFNESS(1:nnz)` and `MASS(1:nnz)` : values on the **same pattern**.

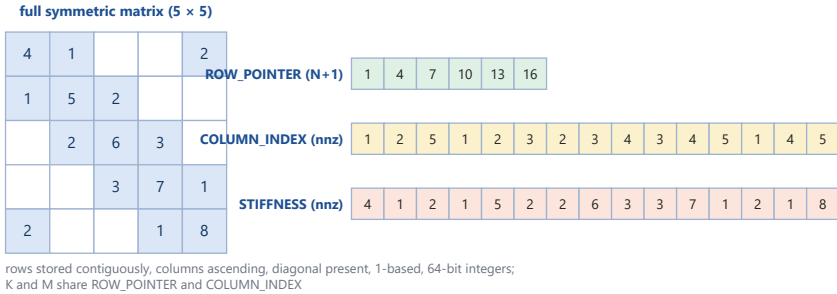


Figure 10.2. Compressed sparse row storage of a small matrix.

The full symmetric matrix is stored (not only the triangle) so that matrix–vector products are simple loops. The direct solver receives the upper triangle, extracted internally.

10.3 The static problem

After the elimination of Section 9, the system is symmetric positive definite (SPD):

$$Kq = F. \quad (10.2)$$

SPD guarantees that the Cholesky factorisation $K = LL^T$ exists and is stable. The program uses the **PARDISO** sparse direct solver of the Intel oneMKL, which applies a fill-reducing reordering, a supernodal numerical factorisation and forward/backward substitution. The matrix type is 2 (real symmetric positive definite). Direct solution is chosen over iterative methods because the matrices of refined CUF models are strongly ill-conditioned (large ratios of stiffnesses between section terms and between thin and thick directions) and iterative methods converge slowly or not at all.

10.3.1 Behaviour on ill-conditioned matrices

Very high-order Taylor models give matrices whose smallest eigenvalues are of the order of the rounding error. The Cholesky factorisation may then meet a non-positive pivot. In this case PARDISO reports the error code `-4` and the program

1. releases the factorisation,
2. retries with the **symmetric indefinite** factorisation (`mtype -2`, Bunch–Kaufman pivoting with pivot perturbation),
3. records the warning *"Cholesky failed (ill-conditioned matrix)"*.

If even that fails the message *"ZERO PIVOT: SINGULAR MATRIX, CHECK SUPPORTS"* is given. The usual causes are: missing or insufficient supports (rigid-body motion left), disconnected parts that are not supported, H8 with MITC (Chapter 8) and a nearly singular Taylor basis (Chapter 4).

10.3.2 Accuracy

For a well-posed problem the solution satisfies the system to about $\epsilon_{mach}\kappa(K)$, where κ is the condition number. High-order TE models and thin structures have large κ ; the *relative displacement error* can then reach 10^{-5} for physically irrelevant reasons. This is one of the reasons why comparisons with reference results use tolerances larger than machine precision.

10.4 Recovery of the solution

The solution vector contains the displacement coefficients in the global frame in the field-major order. The displacement of any point of the structure follows from (4.2); strains and stresses from the strain operator (Chapter 12).

Free vibration

11.1 The generalised eigenvalue problem

After removing the constrained DOFs the free vibration problem is

$$K\phi = \lambda M\phi, \quad \lambda = \omega^2, \quad f = \frac{\omega}{2\pi}, \quad (11.1)$$

with K symmetric positive (semi-)definite and M symmetric positive definite. The eigenvectors can be chosen **M -orthonormal**,

$$\phi_i^T M \phi_j = \delta_{ij}, \quad \phi_i^T K \phi_j = \lambda_i \delta_{ij}. \quad (11.2)$$

The frequencies requested are the lowest ones. A dense solution of (11.1) costs $O(n^3)$ operations and is possible only for small problems; refined CUF models have 10^4 – 10^6 DOF, so a **sparse eigensolver** is required.

11.2 Shift-and-invert Lanczos

The lowest eigenvalues are the *extreme* eigenvalues of the spectral transformation

$$A = (K - \sigma M)^{-1} M, \quad \mu = \frac{1}{\lambda - \sigma}, \quad (11.3)$$

whose eigenvectors are those of the original problem and whose **largest** eigenvalues μ correspond to the eigenvalues λ **closest to the shift** σ . With $\sigma = 0$ (the default) the largest μ are the lowest frequencies. The operator A is self-adjoint in the M inner product, hence the **Lanczos** method applies; its implicitly restarted version is implemented in the **ARPACK** library (routines `dsaupd`, `dseupd`), used in *mode 3* (shift-invert for $Kx = \lambda Mx$ with M in the inner product, `BMAT='G'`).

ARPACK works with *reverse communication*: it never sees the matrices; it returns control with a request, the calling program performs the requested product and calls it again. For mode 3 the requests are

Request (IDO)	Operation
−1 or 1	$y \leftarrow (K - \sigma M)^{-1} Mx$ (a sparse product followed by a back-substitution with the factors)
2	$y \leftarrow Mx$
99	converged / finished

The factorisation of $K - \sigma M$ is computed **once** with PARDISO; each iteration costs one forward-backward substitution and two sparse products. The number of Lanczos vectors is $\max(2 n_{ev} + 1, n_{ev} + 10)$, limited by the problem size.

11.3 Dense fallback

When the free problem has at most 400 DOF or when almost all eigenvalues are requested ($n_{ev} \geq n - 2$) ARPACK is not appropriate (the subspace would be the whole space). The program then builds the dense matrices and solves (11.1) with the LAPACK routine `dsygv`.

11.4 Output and checks

- The eigenvalues are sorted in ascending order and the modes returned ***M*-normalised**.
- The frequency of each mode is $f = \sqrt{\lambda} / 2\pi$ in hertz (negative eigenvalues, which can only be rounding noise of rigid-body modes, are clipped to zero with a warning).
- The relative residual $\|K\phi - \lambda M\phi\| / \|K\phi\|$ is computed for every mode (with the reduced matrices) and written in `ARPACK_SOLVER_INFO.dat`; the console reports the maximum. Values below 10^{-6} indicate converged modes.
- ARPACK is run with machine-precision tolerance and at most 500 restart iterations.

WARNING

If the supports leave rigid-body modes, K is singular and the factorisation of $K - \sigma M$ with $\sigma = 0$ fails. Constrain the structure sufficiently or ask for modes with a non-zero shift (not exposed in the current input format).

11.5 Frequency of a cantilever beam as a sanity check

For an Euler–Bernoulli cantilever of length L , bending stiffness EI and mass per unit length ρA , the first bending frequency is

$$f_1 = \frac{(1.8751)^2}{2\pi} \sqrt{\frac{EI}{\rho AL^4}}. \quad (11.4)$$

A refined CUF model gives a slightly lower value because it includes shear and section deformation. The ratio becomes closer to one for slender beams (Chapter 13 shows an example).

11.6 Participation and mode shapes

Mode shapes are stored as displacement coefficients in the global numbering and expanded to displacement fields on the output grid. In the output files each mode appears as a displacement field with *M*-normalised amplitude; the visual scale is arbitrary and should be set in the post-processor (ParaView, GMSH).

Recovery of displacements, strains and stresses

The solver delivers coefficients; engineering results are values of displacement, strain and stress at **points** or on a **grid**. This chapter explains how they are recovered.

12.1 Locating a point

Given a physical point X_t (a `PNT` request) the program must find the structural element, the sub-element of the expansion mesh and the natural coordinates (ξ_s, ξ_e) that map to it. The map is

$$X(\xi_s, \xi_e) = \sum_i N_i(\xi_s) X_i + R^T \sum_a M_a(\xi_e) c_a, \quad (12.1)$$

where c_a are the local coordinates of the nodes of the expansion sub-element and M_a their shape functions. The search is done in two steps:

1. **Rejection.** Elements whose bounding box (of the structural nodes enlarged by the section extent) does not contain the target are skipped.
2. **Newton iteration.** For every surviving (element, sub-element) pair solve $X(\xi) - X_t = 0$ with the Jacobian of the map, from the centre of the reference domain. The point is accepted when the iteration converges and the natural coordinates are inside the reference element within a small tolerance.

A point that lies outside every element gives a row of `NaN` in `POST_POINT.dat` and a warning.

NOTE

The baseline program locates points only approximately: the evaluated position differs from the requested one by 10^{-7} or more. `MUL2_NEW` reports both, `x_EVAL` (where the field was really evaluated) and `x_REQ` (the request).

12.2 Displacement

At the located point the displacement is the sum (4.2) over the nodes of the element and the terms of the expansion, with the shape functions evaluated at ξ_s and the expansion functions at ξ_e (Lagrange) or at the section coordinates (Taylor). The result is in the **global frame**.

12.3 Strain and stress

The strain column of every DOF is rebuilt from the scalar basis exactly as in the assembly, with the shape and expansion gradients at the point, and the strain is

$$\epsilon_{el} = \sum_I b_I q_I,$$

in the element frame (including the **MITC interpolation of the tied rows** if MITC is active for the element), followed by the stress

$$\sigma_{el} = C_{el} \epsilon_{el}$$

with the stiffness of the lamination of the sub-element. For `GLB` output both are rotated to the global frame: $\epsilon_{gl} = T_{\epsilon}^{-1} \epsilon_{el}$ and similarly for the stress with the stress transformation (Chapter 5).

WARNING

Strains and stresses are computed from the *displacement field* at the point. They are discontinuous between elements and sub-elements (stresses between plies of a laminate in particular) because the displacement approximation is only C^0 . No smoothing or stress-recovery by superconvergent patches is applied. Interlaminar transverse stresses obtained this way are generally less accurate than the in-plane ones; refine the expansion and the mesh to check convergence.

12.4 Output grids

The `PARA` and `GMSH` requests write the fields on a grid that samples every element and sub-element. Each element is divided into $r \times s \times t$ cells (the nine integers of the request give r, s, t separately for beam, plate and solid elements), and every cell has 8, 20 or 27 nodes at natural positions, ordered with ξ outermost and v innermost. The natural coordinates are then split into a structural part and an expansion part:

Table 12.1. Splitting of the natural triple into structural and expansion coordinates.

Element	Structural	Expansion
beam	η (along the axis)	(ξ, v) over the section
plate	(ξ, η) in the surface	v through the thickness
solid	(ξ, η, v)	none

Triangular parents are reached by the collapsed map $a = (1 + \xi) / 2$, $b = (1 + \eta) / 2 (1 - a)$. Quadratic cells (20 or 27 nodes) are written as VTK quadratic hexahedra (cell type 25; only the 20 serendipity nodes are used) and as GMSH second-order elements; linear cells (8 nodes) as VTK type 12. Because every sub-element of the section is sampled, a laminated beam shows each ply as a layer of cells and the discontinuity of stress between plies is visible in the plot.

12.5 Quantities written

At every grid node the program evaluates and writes the displacement vector; the six components of strain and of stress (in the requested frame) and the von Mises stress

$$\sigma_{VM} = \sqrt{\frac{12}{[(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\sigma_{xz}^2 + \sigma_{yz}^2 + \sigma_{xy}^2)]}}. \quad (12.2)$$

For modal analysis the output contains the mode shapes (displacements only), one field per requested mode.

Verification and performance

A finite-element program is trusted only to the extent that it has been compared with independent results. This chapter describes the checks built into the test suite and the measurements that show what accuracy and speed to expect.

13.1 Strategy

1. **Patch tests** (exactness on fields the element must represent).
2. **Comparison with a frozen reference program** (the baseline) on golden inputs and on inputs generated by modifying them.
3. **Internal consistency tests**: two different implementations of the same integral give the same matrices; the result does not depend on the number of threads.
4. **Comparison with closed-form solutions** of beam theory.
5. **Benchmarks** at 100 000 DOF for time and memory.
6. **One suite per later feature** (hierarchical expansions, reduced integration, multiphysics, dynamics, buckling, nonlinear statics, curved beams and shells), each against a closed form or an equivalent model, and a complete aircraft as a robustness check.

13.2 Patch tests

A *patch test* checks that an element can reproduce a field it must be able to represent. With $\nu = 0$ (so that the section does not need Poisson contraction) a uniform axial strain ϵ_0 , $v = \epsilon_0 y$, must be reproduced exactly by every model, for any distortion of the mesh. The program runs this test for beams (B4, Lagrange section, with and without MITC), plates (Q9 with and without MITC) and solids (H8). The error is at the level of rounding (10^{-9} relative or better).

13.3 Comparison with the baseline program

The frozen program (MUL2_OPTIMIZED) has been run on golden inputs (a laminated B4 beam, a Q9 plate, a modal beam) and on inputs obtained by modifying them. The new program reproduces the results as follows.

Table 13.1. Agreement with the baseline (maximum relative difference).

Case	Quantity	Agreement
Laminated beam, B4, LE, MITC	displacements	10^{-7}
same	strains, stresses at points	$2 \cdot 10^{-7}$
Q9 plate, LE through the thickness	all fields	10^{-8}
Mixed model B4 + Q9 + H27, NONE and MITC	all fields	10^{-15}
B3 beam, TE2, MITC	all fields	10^{-6}
Q4 and Q9×4 plates, MITC	all fields	10^{-6} – 10^{-9}
H8 block	all fields	10^{-9}
Modal beam (20 modes)	frequencies	$3 \cdot 10^{-9}$, MAC = 1.000000

Case	Quantity	Agreement
Modal plate, modal TE2 beam	frequencies	$4 \cdot 10^{-6}$

The differences are explained by the penalty used by the baseline for constraints (10^{-7} in displacements, $4 \cdot 10^{-6}$ in frequencies) and by two known approximations of the baseline post-processing, which MUL2_NEW intentionally does not copy:

- the baseline uses, at the tying points of MITC, the Jacobian of the evaluation point instead of that of the tying point (the tied shear strains then differ by $\approx 1\%$ only when the nodes are not uniformly spaced);
- the baseline produces wrong transverse shear strains for plates without MITC.

13.4 Internal consistency

- **Two kernels.** The separable kernel and the point-by-point kernel give the same stiffness and mass matrices to 10^{-15} (dumps of the global matrices compared entry by entry on plate, mixed and solid cases).
- **Thread independence.** The results with one and four threads are *bit-identical*; the assembly scatters in a fixed order.
- **Symmetry.** The element matrices are symmetric by construction; the rigid-body modes of an unconstrained element have zero energy.

13.5 Cantilever beam: comparison between models

A square-section cantilever ($L = 1$ m, 0.1×0.1 m, $E = 70$ GPa, $\nu = 0.3$, $\rho = 2700$ kg/m³) is clamped at $y = 0$ and loaded by $P = 1000$ N at the free end. Beam theory gives, with $I = bh^3 / 12$,

$$\delta_{EB} = \frac{PL^3}{3EI} = 5.714 \times 10^{-4} \text{ m}, \quad \delta_T = \delta_{EB} + \frac{PL}{\kappa GA} = 5.759 \times 10^{-4} \text{ m} \quad (\kappa = 5/6),$$

and from (11.4) the first bending frequency $f_1 = 82.25$ Hz. Five B4 elements are used along the axis.

Table 13.2. Tip deflection and first bending frequency of the cantilever.

Model	DOF	w_{tip} [m]	vs. δ_T	f_1 [Hz]
Euler–Bernoulli / Timoshenko theory	–	$5.714 \times 10^{-4} / 5.759 \times 10^{-4}$	–	82.25
TE1	144	4.282×10^{-4}	–25.6 %	94.61
TE2	288	5.674×10^{-4}	–1.5 %	82.43
TE4	720	5.685×10^{-4}	–1.3 %	82.32
LE, one Q9 section	1 200	5.689×10^{-4}	–1.2 %	82.33
Plate strip (Q9, thickness LE)	459	5.666×10^{-4}	–1.6 %	82.48
Solid (H27)	459	5.663×10^{-4}	–1.7 %	82.53

The comments the table suggests are general:

- **TE1 is far too stiff** (–26 %): it suffers from *Poisson locking*. Under bending the axial strain varies linearly over the section and, through Poisson's ratio, so should the transverse strains; with a linear section kinematics those strains are constant over the section, so the transverse deformation cannot follow and the section is artificially stiffened. TE2 is the lowest useful order.

- All models above TE1 agree within about 0.5 % among themselves; their difference from beam theory (≈ 1.3 %) is the effect of the *three-dimensional clamping*: a perfectly rigid support prevents the Poisson contraction at the root and stiffens the beam slightly, which beam theory neglects.
- The frequencies of the refined models are close to the beam-theory value (82.25 Hz) and approach it from above, as expected for a displacement-based discretisation of a stiffer clamped model.

13.6 Verification of the later features

Every feature added after the first release has its own ctest suite (ctest -C Release: 14 suites, all green; the curved, nonlinear and thermal ones are also run in Debug with /check:all). The tolerances are those of the scripts; the numbers are the ones measured by the last run. The reference is always an independent one: a closed-form solution, an equivalent model of another family, or the same model with another formulation.

13.6.1 Hierarchical expansions (Chapter 14)

Table 13.3. Hierarchical (HLE) elements.

Check	Result
HQ4 $p = 1$ against Q4, static and modal	$2 \cdot 10^{-12}$ / identical
HB $p = 2, 3$ against B3, B4, static and modal, also per ply	$\leq 1.4 \cdot 10^{-12}$
shared side traversed in opposite directions, orders 4 3 and 4 2	$1.7 \cdot 10^{-11}$
fast kernel against reference kernel	$1.7 \cdot 10^{-11}$
C^0 jump on an interface between different orders	$5.7 \cdot 10^{-10}$
Ritz bounds $p2 \leq (4 2) \leq (4 3) \leq p4$ and monotone p -refinement	satisfied
HQ4 $p = 4, 6$ against TE4, TE6	$1.4 \cdot 10^{-4}, 2.6 \cdot 10^{-5}$
circular tube f_1 against Euler-Bernoulli; $p = 4$ against $p = 6$	-0.64 %; $6 \cdot 10^{-5}$

13.6.2 Reduced and selective integration, locking

Table 13.4. Shear modes (REDI, SELI).

Check	Result
uniform extension (beam, plate, solid), REDI and SELI	exact ($\leq 10^{-9}$)
pure bending of the beam	exact ($1.7 \cdot 10^{-11}$)
slender B2 beam, tip/theory: full integration, REDI, SELI	0.036, 0.998, 0.977
slender Q4 plate: full integration, SELI	0.068, 0.988
REDI modal f_1 against full integration	$2.6 \cdot 10^{-3}$

13.6.3 Piezoelectric and thermal fields

Table 13.5. Multiphysics.

Check	Result
slab: $d_{33}V$ thickness strain, $d_{31}E$ lateral strain, $E = -V / t$	$5 \cdot 10^{-7}, 2 \cdot 10^{-10}$, exact
stack, short and open circuit frequency (closed form)	$1.7 \cdot 10^{-8}, 2.6 \cdot 10^{-4}$
beam, plate, solid: free expansion $d_{31}EL$ and $d_{33}V / 2$	$\leq 10^{-6}$ for the three families
beam / plate against solid, short-circuit modes	≤ 2 %

Check	Result
pyroelectric slab, closed and open circuit	exact, $5 \cdot 10^{-7}$
free thermal expansion $\alpha \Delta T L$, stress free, beam / plate / solid	$\leq 10^{-7}$
clamped bar $\sigma = -E\alpha \Delta T$	exact
steady conduction (temperature, flux), all families	exact ($\leq 10^{-8}$)
heat power, Q-PLANE , Q-CONV , Q-SUN , exposed areas	$\leq 2 \cdot 10^{-9}$
thermal bimorph tip $\alpha \Delta T L^2 / 2h$ (beam, solid)	1.3 %, 2.3 %
lumped bar $T(t)$ with convection	$1.9 \cdot 10^{-3}$
D-PLANE / T-PLANE times a field	$1.4 \cdot 10^{-9}$ / exact

13.6.4 Dynamics, buckling, nonlinear statics (Chapter 16)

Table 13.6. Time response, frequency response, buckling, nonlinear statics.

Check	Result
Newmark: period of the first mode against 103	$9 \cdot 10^{-4}$
step load: peak = $2 w_{st}$ / mean / period	2.4 % / 0.15 % / $9 \cdot 10^{-5}$
Rayleigh damping, first overshoot $\exp(-\pi \zeta / \sqrt{1-\zeta^2})$	5.3 %
ramp load: final and half-way deflection	0.2 %, 0.25 %
heat conduction transient, insulated end, series solution	$1.3 \cdot 10^{-2}$ (2000 s), $6.7 \cdot 10^{-4}$ (8000 s); Crank-Nicolson $2.4 \cdot 10^{-3}$
thermoelastic damping: damped only with $T_0 \neq 0$	peak ratio 0.935 against 0.9993
frequency response: static limit, resonance, $1 / 2\zeta$, quadrature	exact, exact, 0.969, -10^{-3}
buckling: cantilever beam first factor against Euler	$6 \cdot 10^{-4}$ (solid: $9.7 \cdot 10^{-3}$)
buckling: bending planes degenerate; third factor with shear	exact; $6.6 \cdot 10^{-3}$
thermal buckling, clamped bar; critical displacement	4.9 %; 1.8 %
nonlinear: small load equals linear; threads	0; bit-identical
elastica (two loads): w / L , u / L against the closed form	$3 \cdot 10^{-3}$, $4 \cdot 10^{-3}$; $6 \cdot 10^{-4}$, $1.5 \cdot 10^{-3}$
beam-column with $P\text{-}\delta$	$6.9 \cdot 10^{-3}$
solid elastica	1.6 %
Newton iterations per step	4-5
arc length equals load control below the limit load	$3 \cdot 10^{-10}$
snap-through of the shallow arch: limit load, coarse against fine	1.9548 / 1.9574 ($1.3 \cdot 10^{-3}$); the load then drops (lowest -1.69)

13.6.5 Curved beams and shells (Chapter 17)

Table 13.7. General geometry.

Check	Result
straight beam named CB2/CB3/CB4 against B2/B3/B4, with and without MITC	identical (0)
flat plate named S4/S9/S16 against Q4/Q9/Q16, TE and LE, MITC and compatible, rotated ply	identical (0)
free quarter ring: rigid modes; first frequency	six; 4.9037 Hz (10^{-5})

Check	Result
quarter ring, tip force $v = 0$; stress under a couple	$6 \cdot 10^{-5}$; 10^{-3}
pinched cylinder (MacNeal-Harder), $8 \times 8 / 12 \times 12$	0.98 / 1.003 (without tying 0.35)
circular tube, tip load against beam theory	0.990
square tube against beam theory ($I = \frac{23}{4} t$, shear included), 4×12	0.97-1.05 (measured 0.995-1.03)
skin and stringer joined at $TE \ 0$ nodes, $E(tb + A) / L$	$2 \cdot 10^{-5}$
clamped wing box of the aircraft, tip load against $PL^3 / 3EI$	13.9 mm against 13.8 mm
straight CB4 and flat S9 in 105 and 108 against B4 and Q9	identical (0)
45-degree bend, 108, $P = 300$: tip against Simo and Vu-Quoc (40.08, -11.87, -6.96) and against a swept H27 solid	0.3, 0.5, 1.2 %; ≤ 0.9 %
circular tube of S9 shells, 105, against the Euler load of the cantilever	1.1 %, two degenerate planes
square tube (kinked shells), 103, first bending frequency against Euler-Bernoulli	2.0 %

Two defects found *by* this campaign and corrected are worth recording because the tests that pass now would not have caught them earlier. (i) The first-order Taylor term shared as a Cartesian vector at an edge between shells of different normals suppresses the bending slope at the edge: a thin box came out 5 to 10 times too stiff. The earlier square-tube test passed because its beam-theory reference used a wrong moment of inertia ($\frac{76}{23}$ instead of $\frac{23}{4}$); the reference was corrected and the node rotation introduced (see the section on joining elements in the chapter on curved beams and shells). (ii) The nodal director was the one of the first element listed at the node instead of the average inside the feature angle. The wing of the aircraft, whose stiffness is known in closed form, is the check that exposed both.

13.6.6 Selection of the fields and joining by coincidence

Table 13.8. FIELDS record and JOIN COINCIDENT.

Check	Result
FIELDS MECH PIEZO against the same input without the record	identical (0)
FIELDS MECH : the potential is switched off	no electromechanical deformation ($< 10^{-15}$ m)
THERMO without temperature expansion, PIEZO without MECH, STRESS (RMVT, reserved), unknown token	explicit error messages
two beam segments with shifted structural nodes (section meshes shifted back) joined by JOIN COINCIDENT against the beam with a shared node	identical (0)
coincident nodes with different numbers; explicit tolerance	identical (0)
same model without the record	segments not joined (rigid-body motion)
malformed record, negative tolerance	explicit error

13.6.7 Complete aircraft (analysis 103 and 101)

A model of 14 433 nodes and 129 093 degrees of freedom (3 706 S9 shells, 4 CB3 beams, 48 H8 solids joined by shared nodes with node-dependent kinematics) has exactly six rigid-body modes and a clean spectrum of global modes from 0.89 Hz (vertical bending of wings and fuselage). The static solution under

symmetric wing-tip loads is symmetric to $5 \cdot 10^{-5}$ (tip displacement 0.33010 m and 0.33012 m). The run takes 2 minutes (modes) or 27 seconds (static) on the test machine. There is no closed-form reference for the whole aircraft: it is a robustness and performance demonstration, the accuracy of the parts is established by the tables above.

13.7 Performance at 100 000 DOF

Two models of about 100 000 DOF were run with the baseline and with MUL2_NEW (Release build, 16 cores, sequential MKL; one run each).

Table 13.9. Time and peak memory at 100 000 DOF.

Model	Analysis	Baseline	MUL2_NEW
Beam, B4, Taylor 15, 100 776 DOF	static	82.9 s, 12.9 GB	26.4 s, 7.8 GB
Beam, B4, Taylor 15, 100 776 DOF	6 modes	183.7 s, 13.5 GB	80.3 s, 13.9 GB
Beam, B4, 16 Q9 section elements, 99 387 DOF	static	14.1 s, 4.1 GB	6.4 s, 1.4 GB
Beam, B4, 16 Q9 section elements, 99 387 DOF	6 modes	28.1 s, 4.2 GB	14.8 s, 1.6 GB

The assembly of the Taylor-15 model went from 526 s (point-by-point) to 12 s with the separable kernel; the Lagrange model benefits from excluding structural zeros (20 M instead of 120 M non-zeros). The accuracy of the solutions is the same as in Table [Table 13.1](#).

13.8 Known limitations and open points

- H20, T4, T10 and P6 are accepted by the input parser but their shape functions are not implemented: using them gives an explicit error.
- MITC with H8 was reported singular in an early version; the validation campaign found it regular (see the Validation Report).
- Triangular sections (T3) use a one-point rule and are unreliable for beams; use T6 or Q9.
- Very high Taylor orders on small sections are numerically singular unless the section is normalised (Chapter 4).
- Hierarchical expansions exist only for quadrilateral sections and thickness lines (Chapter 14); user-defined ($M < N$) expansions are not available.
- Only point loads are available; distributed loads must be replaced by equivalent nodal point loads.

Hierarchical Legendre expansions

The Taylor expansion (TE) converges slowly for sections with holes or corners, and the Lagrange expansion (LE) refines the section only by adding nodes (h-refinement). The **hierarchical Legendre expansion** (HLE) refines by raising the polynomial order p of each section sub-element (p-refinement), with different orders in different sub-elements. MUL2_NEW follows the formulation of Pagani, de Miguel and Carrera (Comput. Mech., 2017).

14.1 The expansion functions on a quadrilateral

A section sub-element HQ is a quadrilateral with natural coordinates $(r, s) \in [-1, 1]^2$ and vertices $1(-1, -1)$, $2(1, -1)$, $3(1, 1)$, $4(-1, 1)$. Its functions come in three kinds.

1D functions. With L_k the Legendre polynomial of degree k ,

$$\phi_k(x) = \frac{L_k(x) - L_{k-2}(x)}{\sqrt{2(2k-1)}} = \sqrt{\frac{2k-1}{2}} \int_{-1}^x L_{k-1}(t) dt, \quad k \geq 2, \quad (14.1)$$

which vanishes at $x = \pm 1$: it can be added to a linear function without changing the end values.

Vertex modes (bilinear, equal to the Q4 functions):

$$N_\tau = \frac{1}{4}(1 + r_\tau r)(1 + s_\tau s), \quad \tau = 1 \dots 4. \quad (14.2)$$

NOTE

the formula with $(1 - r_\tau r)(1 - s_\tau s)$ that appears in some printed versions gives the function of the *opposite* vertex. The code and the tests use the form above.

Side modes, for $k = 2 \dots p$ (one per side):

$$N_k^1 = \frac{1}{2}(1 - s)\phi_k(r), \quad N_k^2 = \frac{1}{2}(1 + r)\phi_k(s), \quad N_k^3 = \frac{1}{2}(1 + s)\phi_k(r), \quad N_k^4 = \frac{1}{2}(1 - r)\phi_k(s). \quad (14.3)$$

Internal modes (bubbles), for $k \geq 4$: $\phi_i(r)\phi_j(s)$ with $i, j \geq 2$ and $i + j = k$. The set is the *trunk space* of Szabó and Babuška, not the tensor product Q_p : the number of functions is

$$n(p) = 4 + 4(p - 1) + \frac{(p - 2)(p - 3)}{2} \quad (p \geq 4), \quad (14.4)$$

that is 4, 8, 12, 17, 23, 30, 38, 47 for $p = 1 \dots 8$. Order 1 is the bilinear Q4; order 2 is the 8-function serendipity space (**not** Q9). The local order of the modes follows the paper: the four vertices, then for each k the four sides of order k followed (for $k \geq 4$) by the internal modes with $i + j = k$.

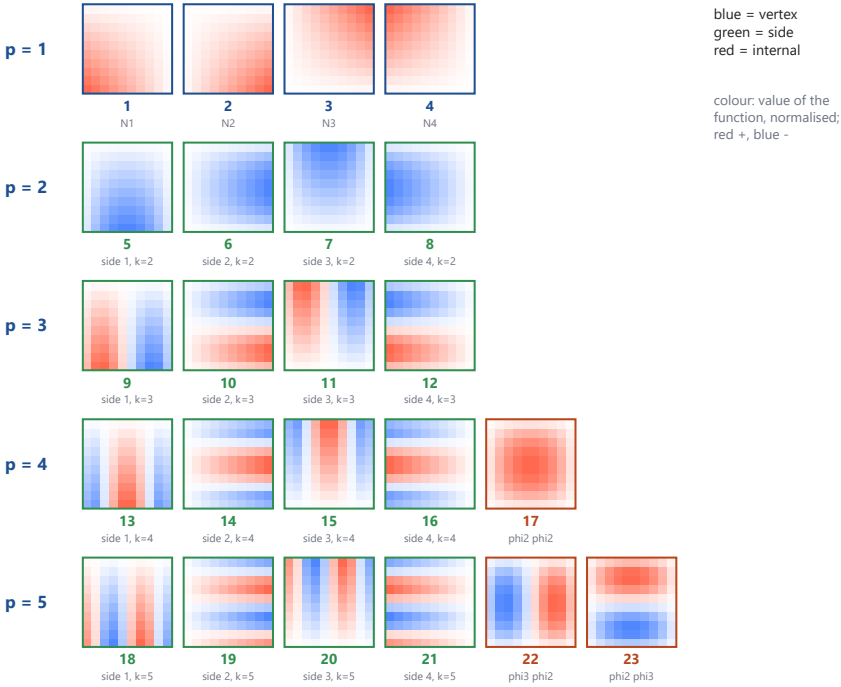


Figure 14.1. Examples of the HLE functions on the reference quadrilateral.

For a line sub-element (HB, used for the thickness of plates) the functions are the two linear vertex functions and ϕ_k , $k = 2 \dots p$. HB of order 2 and 3 span the same spaces as the Lagrange B3 and B4.

14.2 What is a degree of freedom

Only the vertices are nodes of the section mesh. Side and internal modes are *generalised* unknowns attached to a side (shared by the two sub-elements that own it) or to the sub-element. In the field-major numbering the "terms" of an HLE structural node are the vertex terms plus the side and internal terms of the section mesh; they are the same for all the structural nodes of the model.

14.3 Different orders on an interface

Two sub-elements of orders p_1 and p_2 share a side. For the displacement to be continuous (C0) on the side, only the side modes of order $k \leq \min(p_1, p_2)$ are kept: the modes of higher order on the richer side are **omitted** (their trace would not be matched). Internal modes involve no constraint. This is the rule of the paper for modes with a compatible trace and it removes the need for multipoint constraints.

Signs. The side modes are defined with respect to the direction of the side in the element. If the neighbour traverses the shared side in the opposite direction, $\phi_k(-x) = (-1)^k \phi_k(x)$, so the shared unknown is multiplied by $(-1)^k$ in one of the two elements. MUL2_NEW fixes the *global* direction of a side from the

lower to the higher vertex id and applies this sign. The result does not depend on how the sub-elements are numbered (Chapter 13 and the validation report).

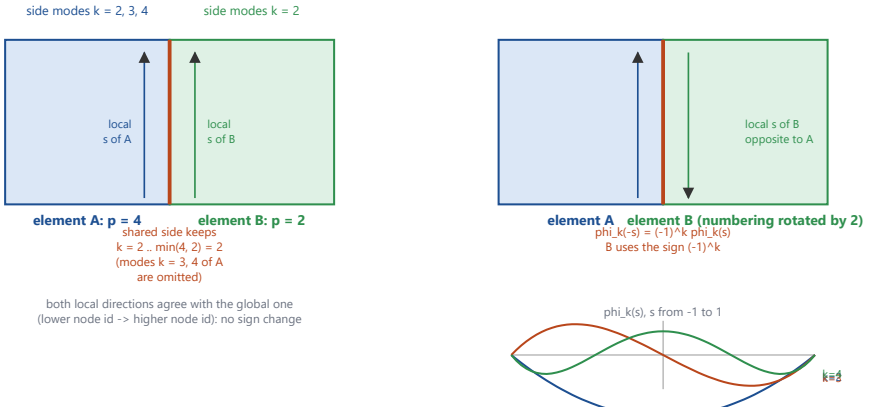


Figure 14.2. Shared side of two sub-elements: orders and signs.

14.4 Curved sides

A side can be an arc of circle through its two vertices and a third point (the optional mid node). The sub-element is then mapped from the reference square with the transfinite (Gordon–Hall) interpolation

$$Q(r, s) = \sum_{\tau=1}^4 N_{\tau} X_{\tau} + \sum_{e=1}^4 \beta_e(r, s) (c_e(t) - l_e(t)), \quad (14.5)$$

where l_e is the straight side, c_e the arc parameterised by its angle, t the natural coordinate along the side, and β_e the linear blending function that is 1 on side e and 0 on the opposite one: $\beta_1 = \frac{1}{2}(1 - s)$, $\beta_2 = \frac{1}{2}(1 + r)$, $\beta_3 = \frac{1}{2}(1 + s)$, $\beta_4 = \frac{1}{2}(1 - r)$. A mid node that lies on the chord means a straight side. The geometry is exact (the area of an annulus sector is integrated to rounding); the expansion functions are **not** the map: the element is not isoparametric, and the physical gradients are obtained with the Jacobian of Q .

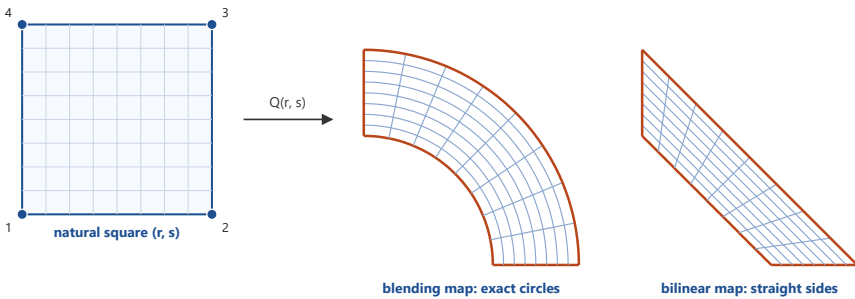


Figure 14.3. The blending map of a curved sub-element.

14.5 Verification of the formulation

Table 14.1. Checks of the HLE (details in the validation report).

Check	Result
HQ4 of order 1 against LE with Q4	identical
HB order 2 / 3 against B3 / B4	10^{-9}
shared side traversed in opposite directions	10^{-9}
C0 jump on the interface, orders 4 3 and 4 2	$6 \cdot 10^{-10}$
compliance of mixed orders	between the uniform bounds (Ritz)
tube with four curved HQ4, $p \geq 4$	within 1.1% of Timoshenko

14.6 Limits

HLE exists for quadrilateral sections (HQ) and for thickness lines (HB) only: no triangles and no solids. Curved sides are circular arcs. A plane constraint that cuts a curved side only constrains the side modes if both end vertices lie on the plane.

Multiphysics: the generalised strain

The Carrera Unified Formulation expands every field as $\phi(x, y, z) = N_i(\xi) F_\tau(x_e) \phi_{it}$, with N_i the structural shape functions and F_τ the expansion over the section. The temperature θ and the electric potential φ are fields like the displacements and use **the same expansion machinery** (Taylor, Lagrange, hierarchical); only the *kinematic operator* and the *constitutive law* differ. This chapter states the formulation in one place; the numerical procedures that use it (time integration, buckling, nonlinearity) are in the next chapter.

Activation. The fields that are solved are selected in the analysis (`FIELDS` record of `ANALYSIS.dat`); the expansions of the fields are defined in the kinematics. The principle of the element is a separate choice: all the elements are displacement-based (PLV); the expansion slots of the transverse stresses $\sigma_{zz}, \sigma_{xz}, \sigma_{yz}$ already exist as explicit fields, so that a mixed (RMVT) element will be an element label, not a change of the data structures.

15.1 1. Generalised strain and constitutive law

At a point the gradients of all the fields are collected in the generalised strain

$$\Gamma = \begin{pmatrix} \epsilon \\ \nabla\varphi \\ \nabla\theta \end{pmatrix} \in \mathbb{R}^{12},$$

with $\epsilon = Bu$ the (engineering) strain, and the generalised stress $\Sigma = \{ \sigma, D, q' \}$ is given by one symmetric matrix

$$\Sigma = M \Gamma - \begin{pmatrix} \beta \theta \\ -p_\epsilon \theta \\ 0 \end{pmatrix}, \quad M = \begin{bmatrix} C & e^T & 0 \\ e & -\epsilon & 0 \\ 0 & 0 & k \end{bmatrix}.$$

In words: $\sigma = C\epsilon + e^T \nabla\varphi - \beta \theta$ with $\beta = C\alpha$ (Hooke's law with piezoelectric and thermal terms), $D = e\epsilon - \epsilon \nabla\varphi + p_\epsilon \theta$ (electric displacement, with $E = -\nabla\varphi$) and $q' = k \nabla\theta$ (the heat flux is $q = -q'$, Fourier's law). C is the stiffness at constant field, e the piezoelectric stress coefficients, ϵ the permittivity at constant strain, α the expansion, k the conductivity and p_ϵ the pyroelectric coefficient at constant strain. All the tensors are given in the material axes and **rotated** into the element frame with the transformation of the engineering strain T :

$$C' = T^T C T, \quad e' = A^T e T, \quad \epsilon' = A^T \epsilon A, \quad k' = A^T k A, \quad \alpha' = T(A^T) \alpha, \quad p' = A^T p.$$

The pyroelectric coefficient that a data sheet gives is measured on a free body, p_o ; the model needs $p_\epsilon = p_o - e \alpha$ (the free expansion contributes $e \alpha \theta$ to D).

Because M contains the blocks side by side, the kernel of Chapter 7 is unchanged: the element matrix is

$$K^e = \int_V B_\Gamma^T M B_\Gamma dV + K_{\text{coup}}^e,$$

where the columns of B_Γ are the strain of a displacement degree of freedom, the gradient of a potential one (rows 7-9) or of a temperature one (rows 10-12). Selective integration only reduces the transverse shear rows of the mechanical block.

15.2 2. Coupling terms and the monolithic system

The temperature enters the mechanical and electric equations through its **value**, not its gradient, so it is outside M :

$$K_{uT}^e = - \int B_u^T \beta N_T dV, \quad K_{\varphi T}^e = + \int (\nabla N_\varphi) \cdot p_\epsilon N_T dV.$$

In the stationary problem the reverse blocks vanish, so the system

$$\begin{bmatrix} K_{uu} & K_{u\varphi} & K_{uT} \\ K_{\varphi u} & -K_{\varphi\varphi} & K_{\varphi T} \\ 0 & 0 & K_{TT} \end{bmatrix} \begin{bmatrix} u \\ \varphi \\ T \end{bmatrix} = \begin{bmatrix} f \\ Q_\varphi \\ q \end{bmatrix}$$

is block triangular and **not symmetric**. It is assembled and solved as one system (a non-symmetric PARDISO factorization), instead of solving the heat problem first and applying a thermal load. K_{TT} is positive definite, the piezoelectric block is indefinite, so the unsymmetric factorization is also the robust choice for thermo-piezoelectric problems.

Why one-way? The heat equation contains the term $T_0 \beta : \epsilon$ (thermoelastic heating); it vanishes in a stationary state because $\dot{\epsilon} = 0$. The transient and harmonic analyses add it (Chapter 16).

15.3 3. Electrodes

A potential prescribed on an electrode is an ordinary constraint of the degrees of freedom on a plane (**V-PLANE**). A **floating** electrode (open circuit) is an equipotential surface with an unknown value: the degrees of freedom of the plane are tied to a master. In algebra, with T the tie matrix, $K \rightarrow T^T K T$ and $f \rightarrow T^T f$ (the charge is the sum on the electrode, zero for an open circuit). The implementation rebuilds the CSR rows with the slaves added to their master, and copies the master value to the slaves after the solution.

Short and open circuit. A modal analysis with the potential degrees of freedom free gives the frequencies of the open-circuit structure, with grounded electrodes those of the short circuit. The potential has no inertia, so its rows are eliminated by static condensation, $K^* = K_{uu} - K_{u\varphi} K_{\varphi\varphi}^{-1} K_{\varphi u}$, before the eigenproblem. For a plate of thickness t polarised across the thickness the fundamental frequency is

$$f_n = \frac{2n-1}{4t} \sqrt{c/\rho}, \quad c_{OC} = c_{33} + \frac{e_{33}^2}{\epsilon_{33}}.$$

the open-circuit modes are stiffer (piezoelectric stiffening), and the code reproduces the closed form within 0.03 %.

15.4 4. Surface loads

A heat flux enters the right-hand side as $\int_\Gamma N_T q d\Gamma$. In CUF a surface is the **product of a part of the structural element and a part of the expansion element**, with two parameters in all:

Element	Surface = structural part × expansion part
beam	axis × edge of the section (lateral skin); end node × whole section
plate	surface × end of the thickness (faces); edge × whole thickness (sides)
solid	face

For a Gauss point of the surface the two natural coordinates give the tangent vectors t_1, t_2 (the structural ones from the node coordinates, the expansion ones rotated by the element frame), the area element is $d\Gamma = |t_1 \times t_2|$ and the normal $n = t_1 \times t_2 / |\cdot|$, oriented away from the centre of the element. A piece is *exposed* when no other piece has the same centre. The loads are

$$q = q_0 F(x), \quad q = aG \max(0, n \cdot s), \quad q = h(T_\infty - T);$$

the convection also adds $h \int N_T N_T^T d\Gamma$ to the conduction matrix. The sun is a plane wave without shadowing.

15.5 5. Spatial fields

A value v of a boundary condition may be multiplied by a function $F(x, y, z)$ from `FIELDS.dat`. For a constraint $vF(x_i)$ is the value of every degree of freedom with the position x_i of its node (Lagrange and vertex hierarchical modes; the Taylor terms of a section take the structural node); for a surface load F is evaluated at the Gauss point.

Time, buckling, frequency and nonlinear analyses

All the analyses of this chapter use the element matrices of the previous chapters and add, per element, the **mass-like matrix** M (the mass $\rho N_i N_j$ on the displacements, the heat capacity $\rho c N_i N_j$ on the temperature, nothing on the potential), the geometric matrix or the nonlinear internal force.

16.1 1. Time response (104)

The semi-discrete equations are, by field,

$$\begin{aligned} M_{uu}\ddot{u} + C\dot{u} + K_{uu}u + K_{u\varphi}\varphi + K_{uT}T &= f(t) \\ K_{\varphi u}u - K_{\varphi\varphi}\varphi + K_{\varphi T}T &= Q_\varphi(t) \\ M_{TT}\ddot{T} + K_{TT}T + C_{Tu}\dot{u} &= q(t) \end{aligned}$$

with the Rayleigh damping $C = \alpha M_{uu} + \beta K_{uu}$ and the thermoelastic heating $C_{Tu} = -T_0 K_{uT}^T$ (it follows from $T_0\beta: \epsilon$; it is active only when the reference temperature T_0 is given). Every load and prescribed value is multiplied by the amplitude $a(t)$.

Newmark for the displacements: with the predictors $\tilde{u} = u_n + \Delta t v_n + \Delta t^2 (\frac{12}{-})\beta a_n$ and $\tilde{v} = v_n + \Delta t(1 - \gamma)a_n$,

$$u_{n+1} = \tilde{u} + \beta \Delta t^2 a_{n+1}, \quad v_{n+1} = \tilde{v} + \gamma \Delta t a_{n+1}.$$

θ -method for the temperature, $\tilde{T} = T_n + \Delta t(1 - \theta)\dot{T}_n$, $T_{n+1} = \tilde{T} + \theta \Delta t \dot{T}_{n+1}$. Substituting in the equations at t_{n+1} gives the effective system, one for all the fields:

$$\left[K + \frac{1}{\beta \Delta t^2} M_u + \frac{1}{\theta \Delta t} M_T + \frac{\gamma}{\beta \Delta t} C_{\text{full}} \right] x_{n+1} = a_{n+1} F + M \left\{ \frac{\tilde{u}}{\beta \Delta t^2}, \frac{\tilde{T}}{\theta \Delta t} \right\} + C_{\text{full}} \left(\frac{\gamma \tilde{u}}{\beta \Delta t} - \tilde{v} \right).$$

For $\gamma = \frac{12}{-}\beta = \frac{14}{-}$ the scheme is unconditionally stable and second-order accurate without algorithmic damping (the numerical period of the first mode of a cantilever differs from the exact one by 10^{-4} at 40 steps per period). The matrix is constant, so it is factored **once**. Prescribed values are imposed by eliminating the constrained columns at every step, which makes a time-varying voltage or temperature an exact boundary condition.

The initial acceleration solves $M_{ff}a_0 = F(0) - Kx_0$ on the free mechanical degrees of freedom.

16.2 2. Frequency response (106)

Under harmonic loads $F e^{i\omega t}$ the response is $x e^{i\omega t}$ with

$$(K - \omega^2 M_u + i\omega (C_{\text{full}} + M_T)) x = F.$$

Writing $x = x_r + ix_i$ and $A = A_r + iA_i$, the complex system is the **real** block system

$$\begin{bmatrix} A_r & -A_i \\ A_i & A_r \end{bmatrix} \begin{bmatrix} x_r \\ x_i \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix},$$

solved by PARDISO on every frequency (the matrix changes with ω , the pattern does not). Near a resonance of a lightly damped structure the response is $\approx \phi \phi^T F / (2\zeta \omega_n^2)$; the code reproduces the modal flexibility fraction of the tip of a cantilever (0.969, theory 0.97) and places the resonance on the frequency of the free-vibration analysis.

16.3 3. Linear buckling (105)

For a structure loaded by λF the stress field is $\lambda \sigma_0$. The potential energy gains a term quadratic in the displacement gradient. In the CUF, with the displacement of a degree of freedom i equal to $N_i F_\tau d_i$ (d_i the unit vector of its component in the element frame) and $g_i = \nabla(N_i F_\tau)$,

$$K_G^{ij} = \int_V (d_i \cdot d_j) g_i^T S_0 g_j dV,$$

where S_0 is the stress tensor of the reference solution (the thermal stress $-\beta\theta$ included). The critical loads are the eigenvalues of

$$(K + \lambda K_G)x = 0.$$

Since K is positive definite the problem is solved as $K_G x = \mu K x$ with $\mu = -1 / \lambda$; the most negative μ give the smallest positive load factors (LAPACK `DSYGV` with the potential degrees of freedom condensed). For a clamped-free beam the code gives P_{cr} within 0.06 % of Euler's $\pi^2 EI / 4L^2$ and, with the shear term, the higher factors of a Timoshenko beam $P_n = P_{E,n} / (1 + P_{E,n} / \kappa GA)$ within 0.7 %.

16.4 4. Geometrically nonlinear statics (108)

The formulation is **total Lagrangian**. With $H = \nabla u$ the displacement gradient, $F = I + H$, the Green strain and the second Piola-Kirchhoff stress are

$$E = \frac{12}{\zeta} H + H^T + H^T H, \quad S = C E - \beta\theta + e^T \nabla \varphi \quad (\text{St. Venant-Kirchhoff}).$$

In the generalised notation the strain is $\Gamma(u) = \{E(u), \nabla \varphi, \nabla \theta\}$ and the variation of E for a degree of freedom i is the linear strain plus the term $\frac{12}{\zeta} H^T d_i \otimes g_i + g_i \otimes H^T d_i$ (in Voigt form, with doubled shears).

The internal force and the tangent matrix of an element are

$$f_{\text{int}}^i = \int B(u)_i^T \Sigma dV, \quad K_T^{ij} = \int B(u)_i^T M B(u)_j dV + \int (d_i \cdot d_j) g_i^T S g_j dV + K_{\text{coup}}^{ij}.$$

The solution is the **Newton-Raphson** iteration

$$K_T(x_k) \Delta x = \lambda F - f_{\text{int}}(x_k), \quad x_{k+1} = x_k + \Delta x,$$

on N increments $\lambda_n = n / N$ of the loads and prescribed values, until $\|R\| \leq \text{tol} \cdot R_{\text{ref}}$ with R_{ref} the largest force seen so far (a self-equilibrated thermal load has no external force, so a reference based on the current iteration would be zero at convergence). The convection of the surface loads is linear and kept as a separate matrix.

Tests: the elastica of a cantilever ($PL^2 / EI = 1$ and 2) within 0.3 % and 0.06 % of the exact solution; a beam-column under $P_{cr} / 2$ within 0.7 % of the amplification $3(\tan u - u) / u^3$; the Green-measure expansion of a heated bar; $\sqrt{1 + 2\alpha \Delta T} - 1$, to 10^{-10} (the thermal strain is a Green strain, so the coupling terms are consistent). Newton converges in 4-5 iterations per step on all of them, the signature of a consistent tangent.

Arc length. To pass limit points the load factor becomes an unknown, $R(x, \lambda) = \lambda F - f_{\text{int}}(x) = 0$, and a constraint on the increment Δu from the last converged state is added (Crisfield, cylindrical): $\Delta u^T W \Delta u = \Delta s^2$, with W selecting the displacement unknowns. Each iteration solves with the same tangent

$$\delta u_R = K_T^{-1} R, \quad \delta u_F = K_T^{-1} F, \quad \Delta u \leftarrow \Delta u + \delta u_R + \delta \lambda \delta u_F,$$

and $\delta \lambda$ is the root of the quadratic obtained by inserting this update into the constraint; of the two roots the one that keeps the direction of the current increment ($\Delta u_{\text{new}}^T W \Delta u_{\text{old}}$ larger) is taken, which prevents turning back along the path. The predictor is $\delta \lambda = \pm \Delta s / \|\delta u_F\|_W$ with the sign of $\delta u_F^T W \Delta u_{\text{prev}}$ (it flips at a limit point, where λ starts to decrease). The step is adapted with $\Delta s \leftarrow \Delta s \sqrt{5 / n_{\text{it}}}$ and halved when Newton fails. The constraint is imposed on the displacements only, so that temperature or potential unknowns (different units) do not enter the norm; prescribed values must be zero because λF is the whole driving term.

Test: a clamped shallow arch (span 10, depth 1, rise 2.5; the nodes lie on a parabola, but every beam element is straight, because curved elements are not yet supported) loaded at the crown; below the limit load the path equals load control to $3 \cdot 10^{-10}$, the limit load ($\lambda \approx 1.96$) found with two very different step sizes agrees within 0.14 %, and the path continues through the descending branch (the load drops below half of the limit load) up to the inverted configuration.

Shear locking. MITC ties the linear part of the strain only; the quadratic part $H^T H$ is not tied. For slender beams and plates with large rotations a fine mesh or reduced integration of the nonlinear terms should be checked.

Curved beams and shells with nodal triads

The elements of the previous chapters have a single frame: the structural geometry is described in the plane (or on the line) of that frame and the expansion lives in the remaining directions. This chapter removes the restriction. A **curved beam** and a **shell** of arbitrary geometry are described by the same kinematics (CUF, Cartesian components) and by a *three-dimensional map* built from nodal triads. Beam and shell share one kernel.

17.1 Geometry: the nodal-triad map

Let ξ be the natural coordinates of the structural element ($d_s = 1$ for a beam, $d_s = 2$ for a shell) and c the position of a point of the expansion mesh in its local axes (section of the beam, thickness of the shell, in lengths). The point of the element is

$$X(\xi, c) = \sum_i N_i(\xi)[X_i + c_1 A_{1i} + c_2 A_{2i} + c_3 A_{3i}],$$

with X_i the node and (A_{1i}, A_{2i}, A_{3i}) its **triad**:

Element	A_1	A_2	A_3
beam	$A_2 \times A_3$	tangent of the axis	reference vector projected on the normal plane
shell	reference vector projected on the tangent plane	$A_3 \times A_1$	director V_i

For a flat plate A_3 is the normal and the map reduces to $X = X_s(\xi) + z n$, the formulation of the plates. For a shell with the exact normals as directors the map is the one of the shell element with nodal directors, $X = \sum N_i(X_i + z V_i)$ with $z = \zeta h / 2$ the physical thickness coordinate; the thickness comes from the expansion mesh, not from a nodal value.

Directors and tangents at a node are the mean of the normals (of the tangents, with the sign aligned) of the elements that share it, *restricted to the elements whose normal is within the feature angle $\theta_f = 40^\circ$ of the normal of the one that is being built*:

$$V_{ie} = \frac{\sum_{e'} n_{e'i} H(n_{e'i} \cdot n_{ei} - \cos \theta_f)}{\|\sum_{e'} \dots\|}, \quad n_{ei} = \frac{a_1 \times a_2}{\|a_1 \times a_2\|} \Big|_{\xi_i},$$

with $a_\alpha = \sum_k \partial_\alpha N_k X_k$ at the natural coordinates of the node. A smooth surface has one continuous director per node (the faces of neighbouring elements coincide, $X^A = X^B$ on the shared side for every c); on a true edge every element keeps its own director. The nodal director of the optional file replaces the mean for the elements within θ_f of it.

Jacobian. The active natural coordinates are the structural ones followed by the active axes of the expansion mesh, $t = (\xi_1, \dots, \xi_{d_s}, c_{k_1}, \dots)$, with $d_s + d_e = 3$. The rows of the Jacobian are

$$g_\alpha = \frac{\partial X}{\partial t_\alpha}, \quad g_\alpha = \sum_i \partial_\alpha N_i (X_i + c_k A_{ki}), \quad g_{d_s+m} = \sum_i N_i A_{k_m i},$$

$J = [g_1; g_2; g_3]$, and the volume element

$$dV = |\det J| \, d\xi \, dc.$$

$g_\alpha = a_\alpha + c b_\alpha$ is linear in the thickness: $J = J_0 + c J_1 = (I + c S) J_0$, and for exact normals $\det(I + c S) = 1 - 2Hc + Kc^2$ (H, K mean and Gauss curvature): the factor $(1 + c / R_1)(1 + c / R_2)$ of the Lamé shells is **included without knowing the radii**, for any curvature. The determinant is not factorised in the implementation: it is computed at every point.

17.2 Strains in the covariant basis

The displacement is $u = \sum_{\tau i} F_\tau(c) N_i(\xi) q_{\tau i}$ with q in global components. With $\varphi = F_\tau N_i$ and e_f the global direction of the component f ,

$$\tilde{\epsilon}_{ab} = \frac{12}{c} g_a \cdot u_{,b} + g_b \cdot u_{,a} = \frac{12}{c} \varphi_{,b} g_{a,f} + \varphi_{,a} g_{b,f}, \quad \gamma = (\tilde{\epsilon}_{11}, \tilde{\epsilon}_{22}, \tilde{\epsilon}_{33}, \tilde{\gamma}_{13}, \tilde{\gamma}_{23}, \tilde{\gamma}_{12}),$$

(the order is the one of the code, xx, yy, zz, xz, yz, xy). The strain of the **material frame** $(e_1, e_2, e_3) = R$ is obtained with $Q_{ma} = e_m \cdot g^a$, $Q = R J^{-1}$:

$$\epsilon_{mn}^L = Q_{ma} Q_{nb} \tilde{\epsilon}_{ab} \Rightarrow \epsilon^L = T(Q) \tilde{\gamma},$$

$T_{rs} = Q_{ma} Q_{na}$ for the normal components and $T_{rs} = \frac{12}{c} Q_{ma} Q_{nb} + Q_{mb} Q_{na}$ for the shears, twice that for the rows with $m \neq n$ (engineering shear). The frame R is built at every point: beam, e_2 the tangent of the axis and e_3 the projected reference vector; shell, e_3 the normal $g_1 \times g_2$ and e_1 the projected reference vector. This is the convention of the plates and of the straight beams, so the cached constitutive matrices of the laminates (ply angle measured from e_1) are used **unchanged**.

Without tying $T\tilde{\gamma}$ is exactly the Cartesian strain of a solid: $B^L = \text{sym}(\nabla \varphi \otimes e_f)$ rotated to the material frame. This is a check of the implementation (the algebra of the kernel gives, to the last digit, the results of the straight beam and of the flat plate, which have another code path).

The potential and the temperature use the Cartesian gradient $R J^{-1} \partial_c \varphi$ and the generalised constitutive matrix of the multiphysics chapter, so the piezoelectric, thermoelastic and pyroelectric couplings of Chapter 15 hold for curved elements (the assembly of K_{uT} and $K_{\phi T}$ uses the same columns).

17.3 Locking and MITC

A shell with compatible strains locks in transverse shear and in membrane; a curved beam in shear and, with quadratic elements, in membrane. The remedy is the interpolation of the assumed strains on the covariant components, **at the same position in the thickness c** :

$$\tilde{\gamma}_{ab}^{AS}(\xi, c) = \sum_m L_m(\xi) \tilde{\gamma}_{ab}(\xi_m, c),$$

with the tying points ξ_m and the Lagrange interpolants L_m of the table ($a = 1 / \sqrt{3}$, $b = \sqrt{3/5}$):

Element	Components	Tying points	L_m
S9 (MITC9)	$\tilde{\epsilon}_{11}, \tilde{\gamma}_{13}$	$\xi \in \{\pm a\}, \eta \in \{-b, 0, b\}$ (6)	linear in ξ , quadratic in η
	$\tilde{\epsilon}_{22}, \tilde{\gamma}_{23}$	$\xi \in \{-b, 0, b\}, \eta \in \{\pm a\}$ (6)	quadratic, linear
	$\tilde{\gamma}_{12}$	$\{\pm a\} \times \{\pm a\}$ (4)	bilinear
S4 (MITC4)	$\tilde{\gamma}_{13}, \tilde{\gamma}_{23}$	$(0, \pm 1), (\pm 1, 0)$	linear
CB2 CB3 CB4	$\tilde{\epsilon}_{11}, \tilde{\gamma}_{12}, \tilde{\gamma}_{13}$	the $n - 1$ Gauss points of the axis	Lagrange of degree $n - 2$
S16	none		

$\mathcal{E}_{33}$ stays compatible. The rows are replaced **before** the rotation to the material frame, with the T of the integration point (not of the tying points). The tying uses g_a and the shape functions of the tying point and the factors F_τ, F_τ, ζ of the integration point. The values of the table are the ones of MITC9 (Bucalem and Bathe) used in the CUF shell papers; the flat s_9 reproduces the MITC of the ordinary plate q_9 , and the same for s_4 and for the beams, to the last digit.

17.4 Integration and matrices

The stiffness is the sum over the points of the combined rule (structural $\times$ expansion), $K^{tsij} = \sum_p W_p B_p^{L,T} M B_{sj}^L$, with

$$W_p = w_\xi w_c | \det J_c |_{\text{exp}} | \det J | ,$$

$w_c | \det J_c |$ the weight and the Jacobian of the expansion mesh (thickness variations are free, every sub-element has its own length). The generalised constitutive matrix M is the cached one (6, 9 or 12 rows). The mass matrix is $M^{tsij} = I_3 \sum_p W_p \rho F_\tau F_s N_i N_j$ (rotary inertia of the thickness is in the terms of F_τ) and the heat capacity is added when the temperature is active.

17.5 Joining elements

The displacements are global components of the nodes, so **continuity is node sharing**. For two shells that share a smooth edge the maps agree for every c (same V_i). At a true edge the elements have different directors $V_a \neq V_b$ at the node and the Taylor term $q_1 = \partial u / \partial c$ has a different meaning in each of them (rotation of one wall, thickness stretch of the other): sharing it as a Cartesian vector is a spurious constraint that suppresses the bending slope at the edge (a thin box came out 5 to 10 times too stiff). The first-order term of an element is therefore written as the rotation ω of the node acting on the director of that element, $q_1^e = \omega \times V^e$, with ω the shared unknown (three components, no multiplier). Both walls see the same rigid rotation, so displacements and rotations are continuous; the thickness stretch of the first-order term is dropped at the node (the second-order term remains). The transformation acts on the columns of the strain and of the basis, so stiffness, mass, loads and recovery use it. Normals exactly opposite at a node (trailing edge) share q_1 with the sign of the thickness axis, which is the continuity of u along the same line. The beam-shell joint uses nodes of order 0 on both sides (terms q_0 only), the one that has the same number of terms in a 1D and a 2D expansion mesh; the stringer is a rod of exact axial stiffness, and since a node of order 0 carries no rotation it must be an isolated node (a line of them across the bending direction locks the shell).

17.6 Recovery

At an arbitrary point the element is located with a Newton iteration on $X(\xi, c)$ (the columns of the Jacobian are g_a and the combination of the nodal triads for the expansion coordinates), the shape and expansion factors are evaluated and the same columns B^L of the stiffness give strain and stress, in the frame of the point.

17.7 Verification

Check	Result
straight beam named CB2 CB3 CB4, flat plate named S4 S9 S16, with and without MITC, TE / LE, rotated ply	equal to the ordinary element to 10^{-9}

Check	Result
free quarter ring, 103	exactly six rigid-body modes; first frequency 4.9037 Hz, converged to 10^{-5}
quarter ring, tip force, $\nu = 0$	energy solution to $6 \cdot 10^{-5}$ (4 B4)
ring, tip couple	$\sigma = Mx / I$ to 10^{-3}
pinched cylinder (MacNeal-Harder), TE 2, LE	0.98 of $1.8248 \cdot 10^{-5}$ with 8×8 per octant; 1.003 with 12×12 ; compatible strains: 0.35 with 6×6
circular and square tube, shared corner nodes	0.990, 0.995 to 1.03 of beam theory

Note on the first-order convergence of the clamped cantilever: with $\nu \neq 0$ the 3D solution has a singularity at the clamp edge and every element of the family converges as h^{-1} (the straight cantilever of the validation suite behaves in the same way); with $\nu = 0$ the curved beam converges at once.

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Numerical tables and references

A.1 Gauss–Legendre rules

Table A.1. Points ξ_p and weights w_p on $[-1, 1]$.

n	Points	Weights
1	0	2
2	± 0.5773502691896258	1, 1
3	$0, \pm 0.7745966692414834$	8 / 9, 5 / 9
4	$\pm 0.3399810435848563, \pm 0.8611363115940526$	0.6521451548625461, 0.3478548451374538
$n > 4$	roots of the Legendre polynomial P_n (Newton iteration, tolerance 10^{-15})	$w = \frac{2}{(1 - \xi^2) [P_n'(\xi)]^2}$

A.2 Taylor terms

Table A.2. Number of terms of the Taylor expansion of order N .

N	beam (section, 2D)	plate (thickness, 1D)
0	1	1
1	3	2
2	6	3
3	10	4
4	15	5
5	21	6
8	45	9
10	66	11
15	136	16

A.3 Voigt order and engineering strains

Table A.3. Voigt convention used everywhere (index, name, meaning).

Index	Strain	Stress	Component
1	ϵ_{xx}	σ_{xx}	normal along x
2	ϵ_{yy}	σ_{yy}	normal along y (beam axis)
3	ϵ_{zz}	σ_{zz}	normal along z
4	γ_{xz}	σ_{xz}	shear xz
5	γ_{yz}	σ_{yz}	shear yz
6	γ_{xy}	σ_{xy}	shear xy

A.4 Glossary

Table A.4. Terms used in the guides.

Term	Meaning
CUF	Carrera Unified Formulation
TE / LE / HLE	Taylor / Lagrange / Hierarchical-Legendre expansion
Expansion mesh	finite element mesh of the section (beam) or thickness (plate) that carries the Lagrange functions and defines the integration sub-domains
Sub-element	one element of an expansion mesh
Structural mesh	the 1D, 2D or 3D mesh along the length (beam), surface (plate) or in space (solid)
Versor	reference vector that fixes the rotation of the element frame about the beam axis or in the plate plane
Lamination	pair (material, two orientation angles) attached to a sub-element
Gauss point (combined)	pair (structural point p , expansion point q)
Nucleus	integral of the pair of basis functions of two DOFs; see Chapter 7
DOF	degree of freedom
MITC	mixed interpolation of tensorial components (assumed strain)
CSR	compressed sparse row storage
SPD	symmetric positive definite
ARPACK	library for large eigenvalue problems (implicitly restarted Lanczos)
PARDISO	direct sparse solver of the oneMKL
Baseline	the frozen reference program (MUL2_OPTIMIZED) used to validate this code

A.5 References

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